Security Lab Solution Document

Automation and Infrastructure as Code (IaC)

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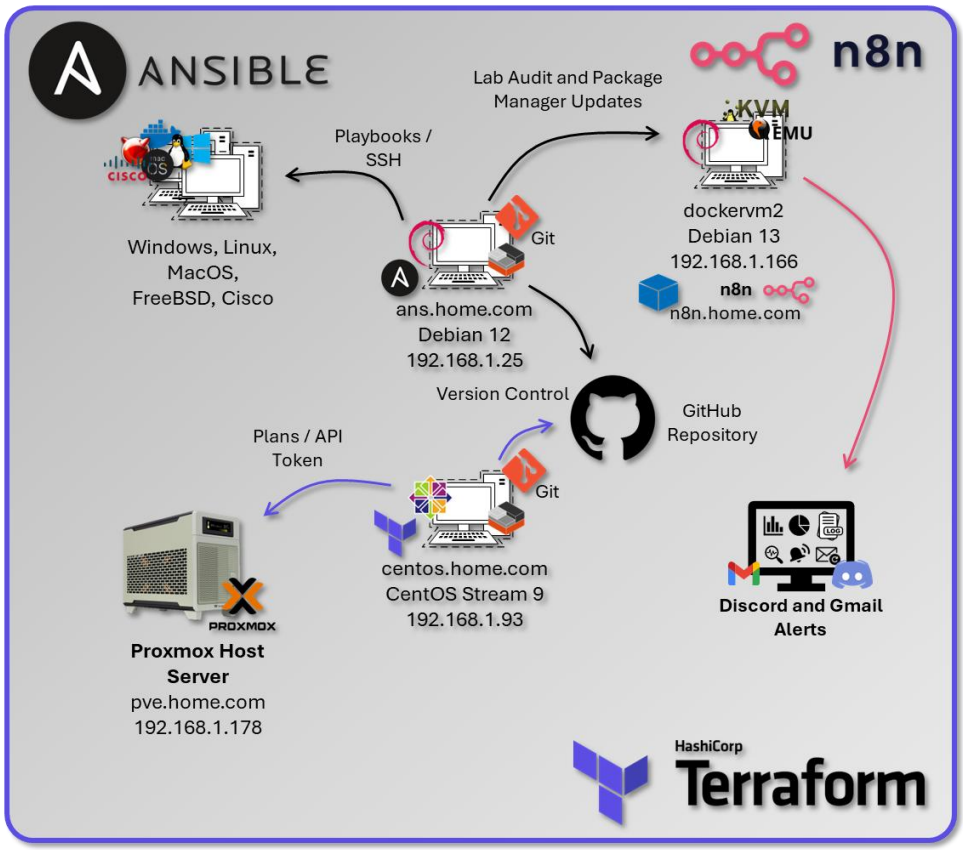
Architecture Overview

The lab implements a comprehensive automation strategy using infrastructure as code principles, configuration management, and workflow orchestration. This approach provides repeatable deployments, consistent configurations, and automated operations across the entire infrastructure stack.



Security Impact

- Configuration drift eliminated through code-driven enforcement
- Human error reduced via automated validation
- Credential exposure prevented through integrated secret management
- Unauthorized changes blocked through Git-based approval workflows
- Disaster recovery enabled through fully reproducible infrastructure
- Audit compliance maintained via immutable version history

Deployment Rationale: Infrastructure as Code eliminates manual configuration drift, reduces human error, and enables rapid disaster recovery. The automation stack transforms a complex 40+ host lab environment into a reproducible, documented system that can be rebuilt from code within hours instead of weeks. This mirrors enterprise DevOps practices and demonstrates proficiency with industry-standard automation tools (Terraform, Ansible, CI/CD pipelines). Enterprise environments leverage IaC to manage hundreds or thousands of servers with consistent security baselines—manual approaches become impossible at scale. This implementation demonstrates understanding of GitOps principles where infrastructure state is declared in version control, changes are peer-reviewed via pull requests, and deployments are automated through CI/CD pipelines.

Architecture Principles Alignment

- **Defense in Depth:** Infrastructure changes peer-reviewed before deployment; Terraform plan reviewed for security impact; Ansible playbooks enforce CIS Benchmarks; automated testing validates security controls before production
- **Secure by Design:** Least-privilege service accounts for automation tools; secrets stored in Ansible Vault/Vaultwarden; SSH keys rotated via automated playbooks; no hardcoded credentials in version control

- **Zero Trust:** Every infrastructure change logged to Git with committer identity; Terraform state files encrypted; API tokens short-lived and rotated; automation execution audited via SIEM

Strategic Value

- **Reduced provisioning time:** VM deployment from 30 minutes (manual) to <5 minutes (Terraform automation)
- **Configuration consistency:** Ansible ensures identical baselines across all hosts (100% CIS Benchmark compliance)
- **Audit trail:** Git commits provide full history of infrastructure changes (who, what, when, why)
- **Disaster recovery:** Entire lab can be rebuilt from GitHub repository (<2 hours full recovery)
- **Learning platform:** Hands-on experience with tools used in production environments (transferable to enterprise roles)
- **Compliance automation:** Security controls codified and version-controlled (auditable evidence of control implementation)

Infrastructure Provisioning with Terraform

Architecture Overview

Terraform manages the complete lifecycle of Proxmox virtual machines and LXC containers using declarative configuration files. The infrastructure is defined as code, version-controlled, and applied through a dedicated automation controller VM running Ubuntu Server.

Security Impact

- Least-privilege automation enforced through Proxmox API tokens
- Infrastructure changes tracked and auditable via Git commit history
- Unauthorized modifications prevented through Terraform's stateful resource management
- Disaster recovery enabled through declarative rebuilds directly from code
- Credential exposure eliminated through encrypted Terraform variables
- Change validation performed prior to execution using terraform plan

Deployment Rationale: Manual VM creation is time-consuming, error-prone, and undocumented—scaling to 40+ VMs/containers requires automation. Terraform demonstrates infrastructure-as-code where VM specifications (CPU, RAM, storage, network) are declared in HCL configuration files, version-controlled, and applied idempotently. This mirrors enterprise infrastructure management (AWS CloudFormation, Azure Resource Manager) where infrastructure is treated as software with code review, testing, and CI/CD deployment. The approach enables rapid environment replication (dev/staging/production), consistent resource configurations, and automated disaster recovery.

Architecture Principles Alignment

- **Defense in Depth:** Terraform state stored remotely with access controls; API tokens scoped to minimal permissions; plan review catches misconfigurations before apply; resource tagging enables compliance auditing
- **Secure by Design:** Proxmox API tokens instead of passwords; Terraform variables encrypted; no secrets in Git repositories; automated validation via pre-commit hooks
- **Zero Trust:** Every Terraform execution logged; state file modifications tracked; API calls authenticated via tokens; infrastructure changes require explicit approval

Deployment Architecture:

Component	Technology	Purpose
Terraform Controller	CentOS LXC (192.168.x.x)	Isolated automation host
Terraform Version	v1.6.x	Infrastructure provisioning engine
Provider	Telmate/proxmox v2.9.x	Proxmox VE API integration
State Backend	Local file, Cloudflare R2	State persistence and locking
Secrets Management	terraform.tfvars (gitignored)	API tokens and credentials

Authentication & Authorization:

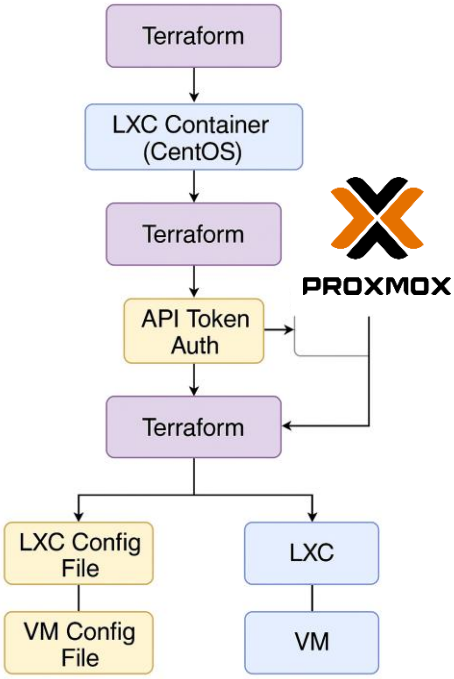
- Proxmox Service Account:
- Username: terraform@pve
- Authentication: API token (non-password, scoped)
- Token ID: terraform-token
- Permission Group: TerraformProvisioners
- Granted Privileges:
    - VM.Allocate (create VMs)
    - VM.Config.* (modify VM settings)
    - VM.PowerMgmt (start/stop/shutdown)
    - Datastore.AllocateSpace (provision storage)
    - SDN.Use (network assignment)
- Principle: Least privilege - cannot modify Proxmox cluster config or other users

Security Controls:

- Secrets management: Promox API token stored in env variable.
- State File Protection: Contains sensitive data, stored with restrictive permissions (0600)
- TLS Verification: pm_tls_insecure = false enforces valid certificate checks
- Separate Workspaces: VM and LXC builds isolated to prevent cross-contamination
- Gitignore Enforcement: terraform.tfvars, *.tfstate excluded from version control

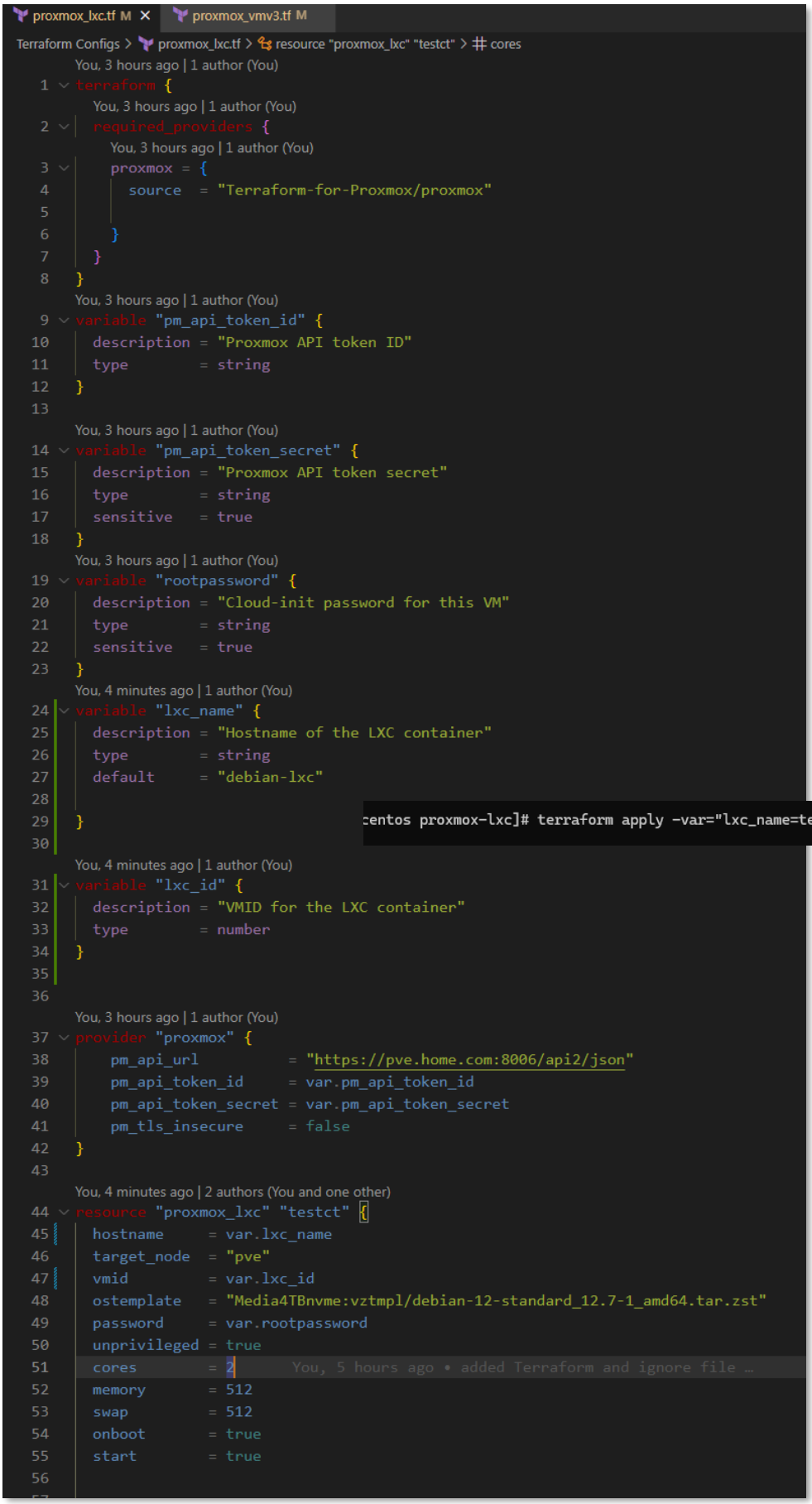
Project Structure:

```
terraform/
├── vm/
│   ├── main.tf            # VM resource definitions
│   ├── variables.tf       # Input variable declarations
│   ├── terraform.tfvars   # Sensitive values (gitignored)
│   ├── outputs.tf         # Return values (IP, VMID)
│   └── .terraform.lock.hcl  # Provider version lock
├── lxc/
│   ├── main.tf            # LXC resource definitions
│   ├── variables.tf       # Input variable declarations
│   ├── terraform.tfvars   # Sensitive values (gitignored)
│   └── outputs.tf         # Return values
└── modules/
    └── common/            # Reusable module components
```



Terraform Configuration Deep Dive

LXC Container Provisioning (main.tf)



Purpose: Deploy unprivileged Debian 12 containers with security isolation and resource limits suitable for containerized workloads.

Key sections:

- **Provider declaration** → Uses Terraform-for-Proxmox/proxmox provider
- **Authentication variables** → API token ID & secret, plus root password for the container
- **Proxmox provider block** → Points to the PVE API over HTTPS with certificate validation (pm_tls_insecure = false)
- **LXC resource** (proxmox_lxc):
  - Hostname & VMID variable arguments provided in the terraform plan command.
  - Template source (ostemplate) from Proxmox storage
  - Unprivileged mode for added security isolation
  - Resource limits: 2 CPU cores, 512 MB RAM + swap
  - Disk allocation: 8 GB on local-lvm storage
  - Networking: virt-bridge to vmbr0 via DHCP
  - Features: nesting enabled (allowing Docker/Kubernetes inside the LXC)
  - Boot & start flags so it powers up with the node

Resource Allocation Strategy:

- CPU: 2 cores (sufficient for most containerized apps)
- Memory: 512MB + 512MB swap (low footprint for density)
- Disk: 8GB (expandable, allocated on local-lvm for performance)
- Network: DHCP with VLAN tagging support via bridge

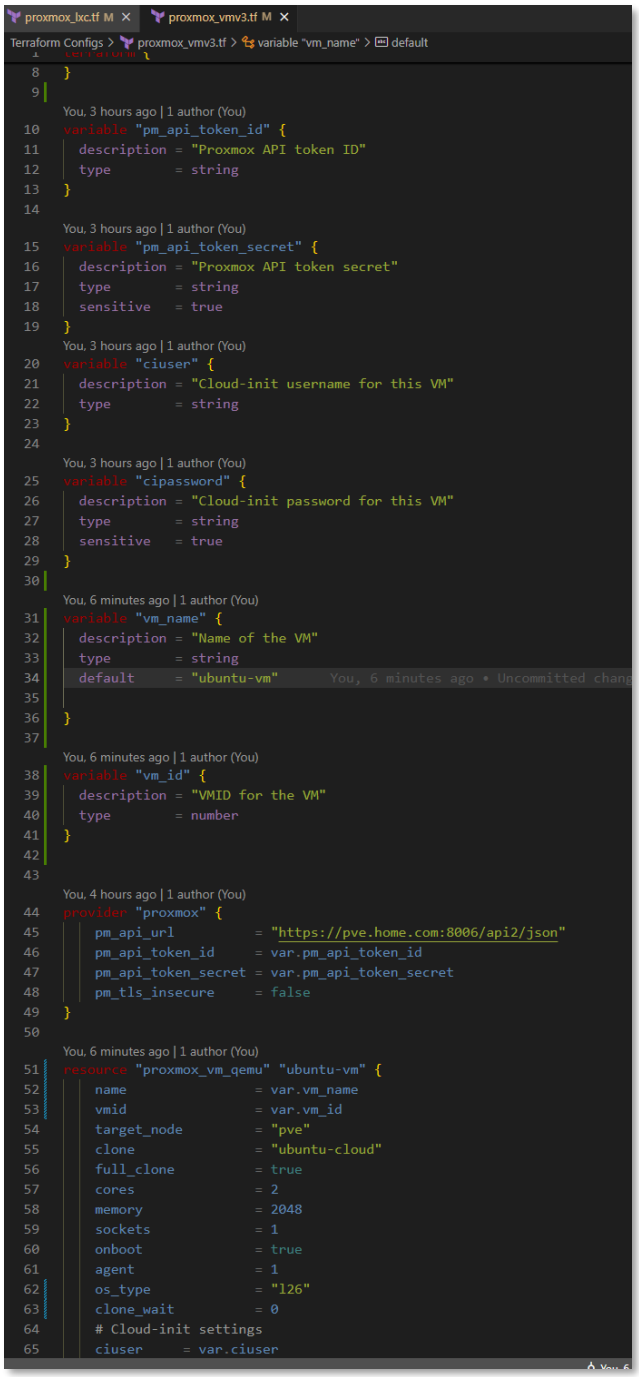
Security Features:

- Unprivileged Containers: UID/GID mapping prevents root escalation to host
- Nesting Enabled: Allows Docker-in-LXC for lab flexibility
- AppArmor Profile: Default Proxmox profile applied
- Resource Limits: Prevent resource exhaustion attacks

VM Provisioning (main.tf - QEMU)

Purpose: Clone cloud-init ready Ubuntu VMs with optimized storage and network configuration for performance and manageability.

Key sections:



- **Provider declaration** → Same Terraform-for-Proxmox provider as above
- **Authentication variables** → API token ID & secret; cloud-init user and password
- **Proxmox provider block** → HTTPS API endpoint with TLS verification
- **VM resource** (proxmox_vm_qemu):
    - Clones from ubuntu-cloud template (must be cloud-init ready)
    - full_clone = true → independent disk image
    - OS type set to l26 → Linux kernel 2.6+ (optimized for modern distros)
    - Hostname & VMID variable arguments provided in the terraform plan command.
    - Boot settings:
        - Boot from scsi0 (OS disk) before ide2 (cloud-init)
        - Force virtio-scsi-single controller for best Linux I/O performance
    - CPU/memory: 2 vCPUs, 2 GB RAM, 1 socket
    - QEMU guest agent enabled for status reporting & IP detection
    - Networking: virtio NIC on vmbr0 with firewall disabled in guest config
    - Cloud-init parameters for provisioning user and password
    - clone_wait = 0 → Terraform doesn't block on post-clone boot readiness

Cloud-Init Integration:

- User Creation: Provisions non-root user via ciuser parameter
- SSH Keys: Injects public keys for passwordless authentication
- Network Config: Configures DHCP or static IP via ipconfig parameters
- Package Updates: Can specify packages to install on first boot

Performance Optimizations:

- VirtIO-SCSI-Single: Modern SCSI controller with single queue (reduces overhead)
- IOThread: Dedicated I/O thread improves disk throughput

- SSD Flag: Enables TRIM for better SSD performance
- VirtIO NIC: Para-virtualized network driver (near-native performance)

Terraform Plan Output

```
[root@centos proxmox-lxc]# terraform apply

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # proxmox_lxc.testct will be created
  + resource "proxmox_lxc" "testct" {
      + arch         = "amd64"
      + cmode        = "tty"
      + console      = true
      + cores        = 2
      + cpulimit     = 0
      + cpuunits     = 1024
      + hostname     = "tf-demo-ct"
      + id           = (known after apply)
      + memory       = 512
      + onboot       = true
      + ostemplate   = "Media4TBnvme:vztmpl/debian-12-standard_12.7-1_amd64.tar.zst"
      + ostype       = (known after apply)
      + password     = (sensitive value)
      + protection   = false
      + start        = true
      + swap         = 512
      + target_node  = "pve"
      + tty          = 2
      + unprivileged = true
      + unused       = (known after apply)
      + vmid         = 223

      + features {
          + fuse    = false
          + keyctl  = false
          + mknod   = false
          + nesting = true
            # (1 unchanged attribute hidden)
        }

      + network {
          + bridge = "vmbr0"
          + hwaddr = (known after apply)
          + id     = (known after apply)
          + ip     = "dhcp"
          + name   = "eth0"
          + tag    = (known after apply)
          + trunks = (known after apply)
          + type   = (known after apply)
        }

      + rootfs {
          + size    = "8G"
          + storage = "local-lvm"
          + volume  = (known after apply)
        }
    }

Plan: 1 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

proxmox_lxc.testct: Creating...
proxmox_lxc.testct: Creation complete after 2s [id=pve/lxc/223]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
```

Terraform Workflow:

Step	Command	Purpose
Initialize	"terraform init"	"Download provider plugins"
Validate	"terraform validate"	"Check syntax errors"
Plan	"terraform plan -var=""hostname=test01"" ..."	"Preview changes"

Apply	"terraform apply -var=""hostname=test01"" ..."	"Execute provisioning"
Destroy	"terraform destroy -var=""vmid=200"""	"Delete resources"
Show State	"terraform show"	"View current state"
Refresh State	"terraform refresh"	"Sync state with reality"

Example Provisioning Command:

```
terraform apply \
  -var="hostname=docker-vm-01" \
  -var="vmid=201" \
  -auto-approve
```

Output: IP address, VMID, MAC address assigned

State Management Strategy:

Current Implementation:

- State File: Local terraform.tfstate in working directory
- Locking: None (single operator environment)
- Backup: Automated weekly backups to Proxmox Backup Server and Synology NAS

Configuration Management with Ansible

Architecture Overview

Ansible provides agentless configuration management through SSH-based automation, enforcing consistent baselines across all managed hosts (Linux, Windows, Cisco, VMware, FreeBSD), managing secrets securely via Ansible Vault, and orchestrating complex multi-host operations using declarative playbooks and Galaxy roles

Security Impact

- Configuration drift eliminated through automated enforcement across 40+ hosts
- SSH hardening applied consistently via dedicated playbooks (PermitRootLogin no, key-only auth)
- Credential exposure prevented through Ansible Vault encryption (AES-256) and vault_* variable pattern
- Audit trail established via Git-based version control
- Manual errors eliminated through playbooks and connectivity pre-tasks
- Multi-platform coverage: Linux, Windows, Cisco IOS, VMware ESXi, FreeBSD (pfSense/OPNsense)

Deployment Rationale: In enterprise environments with 40+ mixed-platform hosts, manual configuration becomes error-prone and time-consuming. This approach reduces configuration time from hours to minutes while ensuring 100% consistency across systems. The lab implementation covers Linux (Debian/RHEL families), Windows Server, Cisco IOS devices, VMware ESXi, and FreeBSD firewalls, demonstrating cross-platform automation capabilities.

Architecture Principles Alignment

- **Defense in Depth:** SSH hardening playbooks disable weak ciphers and enforce key-based auth; firewall rules (UFW/firewalld) deployed uniformly; fail2ban configured consistently; kernel hardening via sysctl parameters; multi-platform audit playbooks provide visibility across the entire infrastructure stack

- **Secure by Design:** Ansible Vault encrypts sensitive variables (vault_ansible_password, vault_root_password); no plaintext credentials in playbooks; SSH keys distributed securely via cloud-init; vault.yml encrypted with AES-256
- **Zero Trust:** Every configuration change logged to Git; playbooks verify current configuration before making changes; connectivity pre-tasks skip unreachable hosts gracefully; password rotation workflow requires explicit vault updates

Setup Overview

Control plane: Ansible running in a Proxmox LXC

- Deployed from a Proxmox Debian-based LXC ("ansible"), using a Python virtual environment.
- SSH auth is key-based via an "ansible" user.
- Handles initial configuration management to achieve a standard template across all hosts.
    - Standard base packages, DNS, PKI and SSH configurations plus user accounts and permissions.

Control Plane Configuration:

Component	Details
Ansible Controller	Debian 12 LXC (192.168.x.x)
Ansible Version	2.16.x (core)
Python Environment	venv isolated (Python 3.11)
Authentication	SSH key-based (ed25519)
Privilege Escalation	sudo (passwordless for ansible user)
Inventory	Dynamic (Proxmox API) + static YAML
Vault Encryption	ansible-vault with AES-256

Inventory and Variable Structure

Inventory Design Principles

The inventory uses a hierarchical group structure with NO host duplication. Each host appears exactly once in a platform-specific group, then aggregated via [group:children] declarations for flexible targeting.

- Host Type Groups: [lxc], [vm], [new]
- OS Family Groups: [debian_lxc], [debian_vm], [redhat_lxc], [redhat_vm]
- OS Aggregate Groups: [debian], [redhat] (children of lxc+vm OS groups)
- Platform Groups: [windows], [cisco], [vmware], [freebsd], [fortigate]
- Function Groups: [monitoring], [dns], [proxy], [pki], [docker], [k3s]
- Meta Groups: [linux] (all managed Linux hosts – excludes [new])

Inventory Code Snippets (hosts.ini)

```
##################################################################
# DEBIAN LXC HOSTS
##################################################################
[debian_lxc]
192.168.1.108      # webserver         web.home.com
192.168.1.136      # Plex              plex.home.com
192.168.1.250      # Pi-hole           pihole.home.com     (Docker)
192.168.1.219      # Wazuh manager     wazuh.home.com

[debian_lxc:vars]
ansible_user=ansible
ansible_password="{{ vault_ansible_password }}"
is_lxc=true

##################################################################
# WINDOWS HOSTS
##################################################################
[windows]
192.168.1.152      # WinServer 2022 / DC    dc01.home.com
192.168.1.142      # WinServer 2025 / DC    dc02.home.com

[windows:vars]
ansible_connection=winrm
ansible_winrm_transport=ntlm
ansible_port=5985
ansible_password="{{ vault_ansible_windows_password }}"

##################################################################
# OS AGGREGATE GROUPS – children only
##################################################################
[debian:children]
debian_lxc
debian_vm

[linux:children]
debian
redhat

##################################################################
# FUNCTION GROUPS
##################################################################
[monitoring]
192.168.1.181      # Uptime Kuma
192.168.1.219      # Wazuh
192.168.1.246      # Grafana
```

Targeting Examples

```
ansible-playbook playbook.yml --limit linux          # All Linux hosts
ansible-playbook playbook.yml --limit lxc            # All LXCs
ansible-playbook playbook.yml --limit debian         # All Debian (LXC + VM)
ansible-playbook playbook.yml --limit debian_lxc     # Debian LXCs only
ansible-playbook playbook.yml --limit lxc --skip-tags reboot  # Skip reboot on LXCs
```

Variable Structure

Variables follow a strict hierarchy: group_vars/all.yml (global defaults) → group_vars/vault.yml (encrypted secrets) → group-specific vars → host_vars/ (host-specific overrides).

Global Variables (group_vars/all.yml)

```
# Ansible connection
ansible_python_interpreter: auto_silent
ansible_user: ansible
ansible_password: "{{ vault_ansible_password }}"
ansible_become: true
ansible_become_method: sudo
ansible_become_password: "{{ vault_ansible_password }}"

# SSH public keys
ansible_ssh_pubkey: "ssh-ed25519 AAAAC3..."
officepc_ssh_pubkey: "ssh-ed25519 AAAAC3..."

# Wazuh agent configuration
wazuh_manager_ip: "192.168.1.219"
wazuh_manager_port: 1514
wazuh_version: "4.14.2-1"

# CheckMK agent configuration
checkmk_server: "http://192.168.1.126:5000"
checkmk_site: "cmk"
checkmk_version: "2.4.0p20-1
```

Vault Structure (group_vars/vault.yml - Encrypted)

```
vault_ansible_password: "<32-char-random-token>"
vault_root_password: "<complex-password>"
vault_paul_password: "<user-password>"
vault_ansible_windows_password: "<windows-password>"
vault_user_paul_password: "<cisco-password>"
vault_enable_password: "<cisco-enable-password>
```

Galaxy Roles and Collections

Ansible Galaxy provides pre-built roles and collections for common tasks. The lab uses officially maintained roles for Wazuh, CheckMK agent deployment Plus support for Windows, VMware, Cisco hosts.

Core Playbooks - Detailed Overview

Playbook 1: Multi-Platform System Audit (sys_audit_n8n.yml)

Purpose: Comprehensive infrastructure audit across Linux, Windows, and FreeBSD hosts with JSON output for n8n workflow processing

Key Features

- Single-play design for all platforms (Linux/Windows/FreeBSD)
- Connectivity pre-checks skip unreachable hosts gracefully
- Platform-specific data collection with conditional blocks
- Consolidated JSON output to /tmp/audit_report.json
- Integration with n8n for HTML report generation and alerting

Data Collected

- System: CPU cores, memory, disk usage %, memory usage %, uptime
- Network: Default gateway, nameservers, listening ports count
- Security: SSH keys count, Windows Defender status, running services
- Software: Kernel version, available updates count

Code Snippet - Connectivity Pre-Tasks

```
pre_tasks:
  - name: Check if host is reachable
    ansible.builtin.wait_for_connection:
      timeout: 5
    ignore_unreachable: true
    ignore_errors: true
    register: connection_check

  - name: Gather facts only for reachable hosts
    ansible.builtin.setup:
    when: connection_check is success

  - name: Skip unreachable host
    ansible.builtin.meta: end_host
    when: connection_check is failed or connection_check is unreachable
```

Code Snippet - Linux Data Collection

```
- name: Collect disk usage (Linux)
  shell: df -h / | tail -n 1 | awk '{print $5}' | sed 's/%//'
  register: disk_usage_pct
  changed_when: false

- name: Collect memory usage (Linux)
  shell: free | grep Mem | awk '{printf "%.0f", ($3/$2) * 100}'
  register: mem_usage_pct
  changed_when: false

- name: Collect package updates (Linux)
  shell: |
    if command -v apt &> /dev/null; then
      apt list --upgradable 2>/dev/null | grep -c upgradable || echo "0"
    elif command -v dnf &> /dev/null; then
      dnf check-update -q 2>/dev/null | grep -v "^$" | wc -l || echo "0"
    fi
  register: updates_available
  changed_when: false
```

Code Snippet - Windows Data Collection

```
- name: Collect disk usage (Windows)
  win_shell: |
    $disk = Get-PSDrive C | Select-Object Used,Free
    [math]::Round(($disk.Used / ($disk.Used + $disk.Free)) * 100, 0)
  register: win_disk_usage

- name: Collect Windows Defender status
  win_shell: (Get-MpComputerStatus).AntivirusEnabled
  register: win_defender_status
  ignore_errors: true
```

Code Snippet - JSON Consolidation

```
- name: Write consolidated JSON report
  copy:
    content: |
      {
        "report_generated": "{{ ansible_date_time.iso8601 }}",
        "report_date": "{{ ansible_date_time.date }}",
        "total_hosts": {{ collected_audit_data | length }},
        "hosts": {{ collected_audit_data | to_nice_json }}
      }
    dest: "/tmp/audit_report.json"
  delegate_to: localhost
  run_once: true
```

Use Cases

- Weekly infrastructure audits via n8n automation
- Pre-maintenance compliance checks
- Capacity planning via disk/memory trending
- Security baseline validation (SSH keys, services, updates)

Playbook 2: User Management (User_mgmt.yml)

Purpose: Centralized user account and credential management across all Linux hosts with Ansible Vault integration

Key Features

- Sets ansible user password from vault_ansible_password
- Sets root password from vault_root_password
- Manages paul user with sudo access (password-based)
- Deploys SSH keys for authorized users
- Verifies ansible key-based auth after password rotation

Code Snippet - Ansible User Password Management

```
- name: Set ansible user password from vault
  tags: ansible_user
  ansible.builtin.user:
    name: ansible
    password: "{{ vault_ansible_password | password_hash('sha512') }}"
    update_password: always

- name: Verify ansible SSH key is still present
  tags: ansible_user, verify
  ansible.posix.authorized_key:
    user: ansible
    key: "{{ ansible_ssh_pubkey }}"
    state: present
```

Code Snippet - Paul User with Sudo Configuration

```
- name: Ensure paul user exists
  ansible.builtin.user:
    name: paul
    shell: /bin/bash
    groups: "{{ 'sudo' if ansible_os_family == 'Debian' else 'wheel' }}"
    append: true
    password: "{{ vault_paul_password | password_hash('sha512') }}"

- name: Deploy sudoers file for paul
  ansible.builtin.copy:
    content: |
      Defaults:paul rootpw
      paul ALL=(ALL) ALL, !/usr/bin/su
    dest: /etc/sudoers.d/paul
    owner: root
    group: root
    mode: '0440'
    validate: '/usr/sbin/visudo -cf %s'
```

Password Rotation Workflow

- Generate 32-char token: openssl rand -base64 24
- Update vault_ansible_password in vault.yml: ansible-vault edit group_vars/vault.yml
- Run playbook: ansible-playbook user_mgmt.yml --tags ansible_user
- Test connectivity: ansible linux -m ping

Use Cases

- Quarterly credential rotation (automated via n8n)
- Emergency password resets
- New host bootstrap user provisioning
- SSH key distribution after key rotation

Playbook 3: Linux Package Updates (update_linux_hosts.yml)

Purpose: Update all Linux hosts with dist-upgrade/dnf update and reboot detection

Key Features

- Debian: apt dist-upgrade with autoremove/autoclean
- RedHat: dnf update with latest packages
- Reboot detection for kernel updates
- Free strategy for parallel execution
- Update summary with reboot status

Code Snippet - Debian Package Updates & Reboot Detection.

```
- name: Update all packages (Debian)
  tags: packages, update
  apt:
    upgrade: dist
    update_cache: true
    cache_valid_time: 3600
    autoremove: true
    autoclean: true
  when: ansible_os_family == "Debian"
  register: apt_update

- name: Check if reboot required (Debian)
  tags: reboot
  stat:
    path: /var/run/reboot-required
  register: reboot_deb
  when: ansible_os_family == "Debian"
```

Use Cases

- Weekly package updates (n8n automation)
- Security patch deployment
- Post-vulnerability scanning remediation
- Compliance maintenance (keep systems current)

Playbook 4: Linux Hardening (linux_hardening.yml)

Purpose: Docker-aware SSH and system hardening with automatic Docker detection

Key Features

- SSH hardening: disable root login, enforce key-only auth
- TCP/Agent forwarding enabled for Docker/DevOps workflows
- Automatic Docker detection preserves IP forwarding
- Safe kernel parameters (sysctl hardening)
- Login banner deployment

Code Snippet - Docker Detection

```
- name: Detect if Docker is installed
  stat:
    path: /usr/bin/docker
  register: docker_check

- name: Set Docker presence fact
  set_fact:
    has_docker: "{{ docker_check.stat.exists }}"

- name: Configure sysctl for Docker hosts
  sysctl:
    name: net.ipv4.ip_forward
    value: '1'
    state: present
    sysctl_set: true
  when: has_docker | bool
```

Code Snippet - SSH Hardening

```
- name: Configure SSH hardening options
  lineinfile:
    path: /etc/ssh/sshd_config
    regexp: '^#?{{ item.key }}\s'
    line: '{{ item.key }} {{ item.value }}'
    state: present
  loop:
    - { key: 'PermitRootLogin', value: 'no' }
    - { key: 'PasswordAuthentication', value: 'no' }
    - { key: 'PubkeyAuthentication', value: 'yes' }
    - { key: 'AllowAgentForwarding', value: 'yes' }
    - { key: 'AllowTcpForwarding', value: 'yes' }
  notify: restart sshd
```

Use Cases

- New host security baseline
- Docker host specialized hardening
- Compliance enforcement (CIS benchmarks)
- Post-compromise hardening

Playbook 5: New Install Baseline (new_install_baseline_roles.yml)

Purpose: Bootstrap new hosts with baseline configuration using Galaxy roles

Roles Used

- wazuh.wazuh_agent (version: 4.14.2)
- Custom bootstrap tasks (user creation, SSH, sudo)

Key Features

- Creates ansible service account with passwordless sudo
- Deploys SSH keys for ansible and paul users
- Configures sudoers with validation
- Installs Wazuh agent and registers with manager
- Installs utility packages (curl, nano, dig, traceroute)
- Enables qemu-guest-agent for Proxmox integration

Code Snippet - User Creation

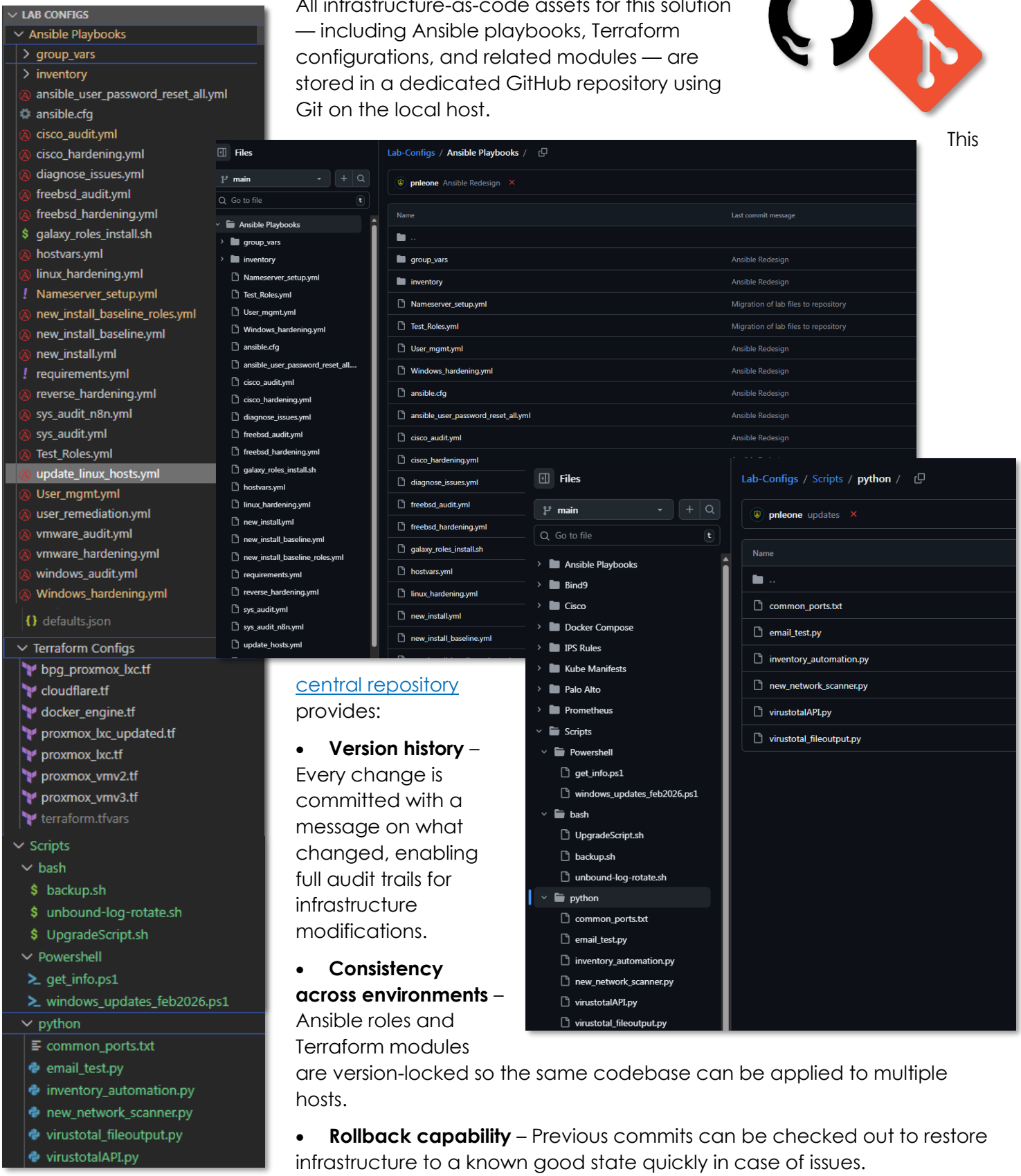
```
- name: Create ansible user
  ansible.builtin.user:
    name: ansible
    shell: /bin/bash
    password: "{{ vault_ansible_password | password_hash('sha512') }}"
    comment: "Ansible Automation Service Account"

- name: Ensure .ssh directory exists
  file:
    path: /home/ansible/.ssh
    state: directory
    owner: ansible
    group: ansible
    mode: '0700'

- name: Add SSH key for ansible user
  ansible.posix.authorized_key:
    user: ansible
    key: "{{ ansible_ssh_pubkey }}"
    state: present

- name: Deploy sudoers file for ansible user (passwordless)
  ansible.builtin.copy:
    content: "ansible ALL=(ALL) NOPASSWD:ALL\n"
    dest: /etc/sudoers.d/ansible
    mode: '0440'
    validate: '/usr/sbin/visudo -cf %s'
```

Use Cases

- Fresh install bootstrap (first playbook to run)
- Wazuh agent deployment across new hosts
- Monitoring stack integration
- Standardized user/SSH configuration

Version Control and GitOps

Version Control Strategy



All infrastructure-as-code assets for this solution — including Ansible playbooks, Terraform configurations, and related modules — are stored in a dedicated GitHub repository using Git on the local host. This central repository provides:

- **Version history** – Every change is committed with a message on what changed, enabling full audit trails for infrastructure modifications.
- **Consistency across environments** – Ansible roles and Terraform modules are version-locked so the same codebase can be applied to multiple hosts.
- **Rollback capability** – Previous commits can be checked out to restore infrastructure to a known good state quickly in case of issues.

Configuration files for hosted and platform services — including YAML, JSON, HTML, CSS, Python, and PowerShell scripts — are stored in a separate repository. This repository is fully integrated with Visual Studio Code, allowing seamless local and remote (via SSH) integration editing.

By keeping both provisioning (Terraform) and configuration management (Ansible) in one repository, and separating hosted/platform service configs into another, the architecture ensures that each layer of the stack is versioned, auditable, and maintainable.

Workflow Automation with n8n

Platform Overview

n8n is a self-hosted, low-code workflow automation platform enabling visual workflow design with conditional logic, error handling, loops, and data transformation. Deployed as a containerized service behind Traefik reverse proxy with Authentik SSO.

Security Impact

- Security operations accelerated through automated SOAR workflows
- Manual triage eliminated via automated alert enrichment
- MTTR reduced through auto-generated remediation playbooks
- Alert fatigue minimized through intelligent deduplication and correlation
- Compliance audit trails automatically documented throughout the incident lifecycle

Deployment Rationale: Security operations generate thousands of events daily, manual triage is unsustainable. n8n demonstrates Security Orchestration, Automation and Response (SOAR) capabilities where vulnerability scans trigger automated ticket creation, threat intelligence enrichment queries multiple APIs, and incident response playbooks execute without human intervention. This mirrors enterprise SOAR platforms (Splunk SOAR, Palo Alto Cortex XSOAR) where analyst efficiency is multiplied through automation.

Architecture Principles Alignment

- **Defense in Depth:** Automated vulnerability remediation workflows execute Ansible playbooks; firewall rule changes logged and reviewed; failed automation triggers manual fallback procedures
- **Secure by Design:** Credentials stored in n8n credential vault (encrypted at rest); webhook endpoints authenticated via Authentik tokens; workflow execution logs forwarded to SIEM
- **Zero Trust:** Every automation action logged with timestamp/user; no hardcoded credentials; API tokens expire and rotate automatically

n8n Configuration:

- Version: n8n v2.7.5 (latest stable)
- Execution Mode: Main process (not queue mode for simplicity)
- Webhook URL: https://n8n.home.com
- TLS Certificate: Step-CA issued, auto-renewed
- Authentication: SSO via Authentik (no local passwords)
- Monitoring: Uptime Kuma
- Notifications: Discord webhooks, smtp relay

Security Controls:

- Credential Encryption: All API tokens encrypted at rest, Ansible automated (ansible-vault)
- Webhook Security: HMAC signature validation on inbound webhooks
- Audit Trail: All workflow executions logged with timestamp and user

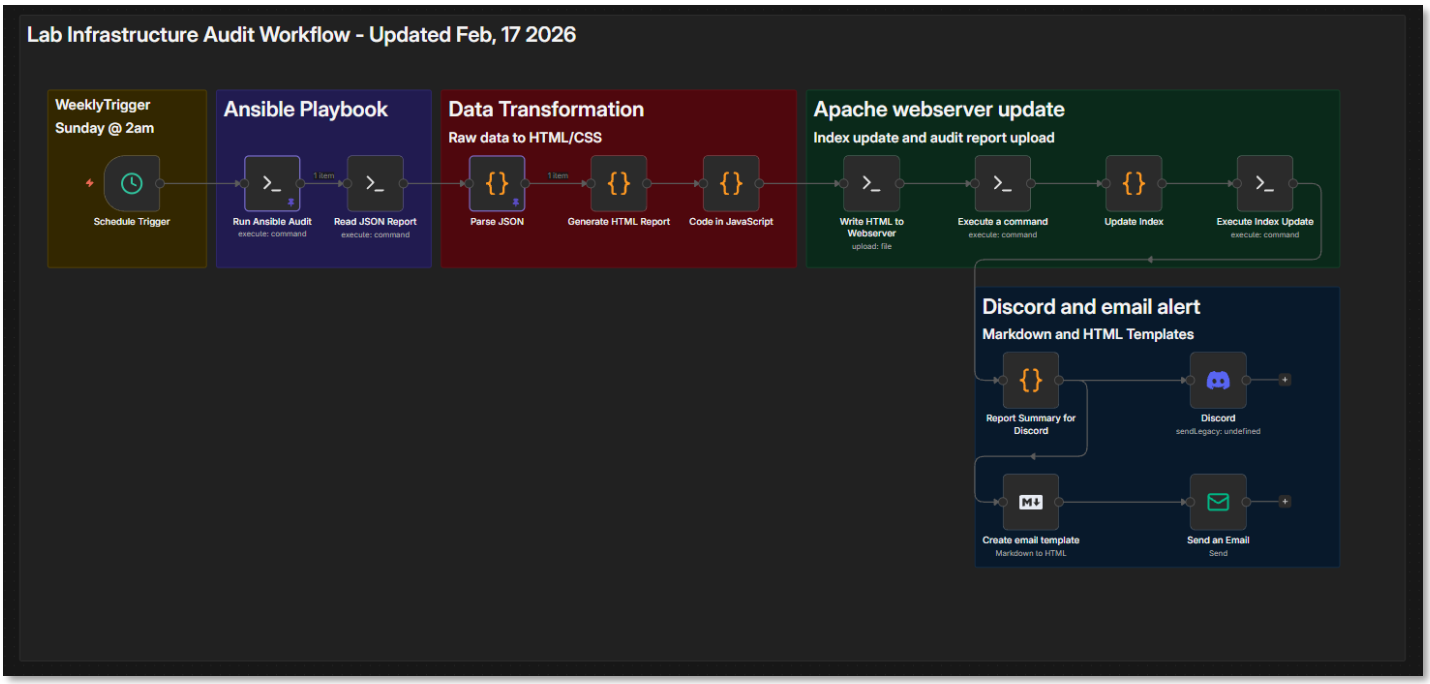
Workflow 1: Lab Infrastructure Audit

Purpose: Automated weekly configuration audit and system updates across lab infrastructure with HTML reporting and dual alerting (Discord + Email)

This workflow runs weekly on Sunday at 2am, executes the Ansible sys_audit_n8n.yml playbook, transforms the JSON output into a styled HTML report, deploys it to the Apache webserver, and sends notifications via Discord and email.

Workflow Summary

- Scheduled Execution: Triggered weekly via n8n's Cron node (Sunday 2 AM)
- Ansible Playbook: Executes sys_audit_n8n.yml via SSH to collect system metrics
- Data Transformation: Parses JSON and generates HTML report with CSS styling
- Apache Upload: Deploys HTML to /var/www/html/ and updates index page
- Discord Alert: Sends formatted notification with report link and summary stats
- Email Alert: Sends HTML email template with audit summary



Workflow Nodes

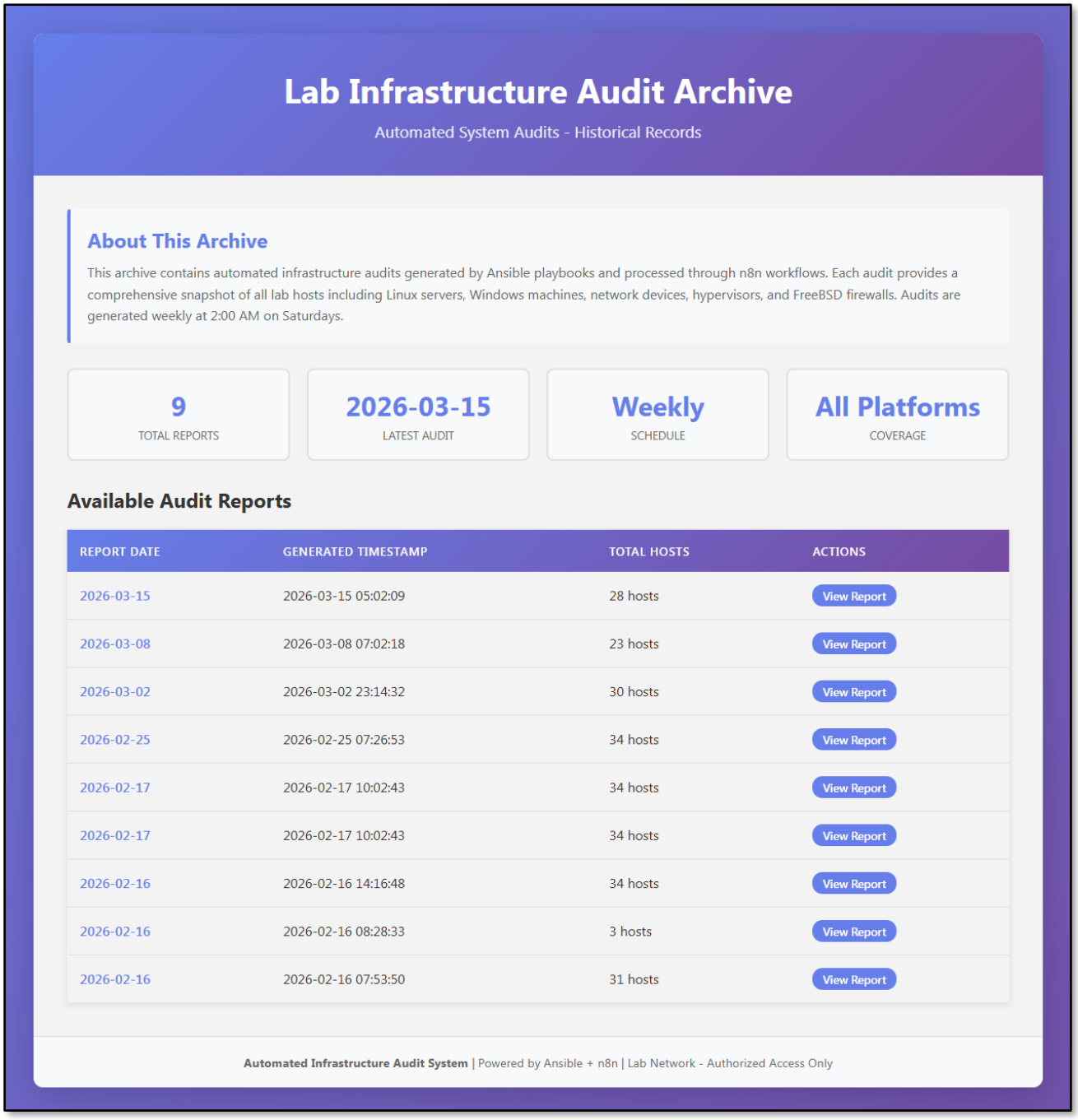
Node Type	Configuration	Purpose
Schedule Trigger	Cron: 0 2 * * 0 (Sunday 2 AM)	Weekly execution
SSH Node 1	ansible-playbook sys_audit_n8n.yml	Run audit playbook
SSH Node 2	cat /tmp/audit_report.json	Read JSON output
Code Node 1	Parse JSON from stdout	Extract audit data
Code Node 2	Generate HTML with CSS	Create styled report
SSH Node 3	Write HTML to webserver	Deploy to Apache
SSH Node 4	Update index.html	Add report to index
Code Node 3	Generate Discord markdown	Format alert message

Discord Webhook	POST to webhook URL	Send Discord notification
Code Node 4	Generate email HTML	Format email template
Email Node	SMTP send	Send email notification

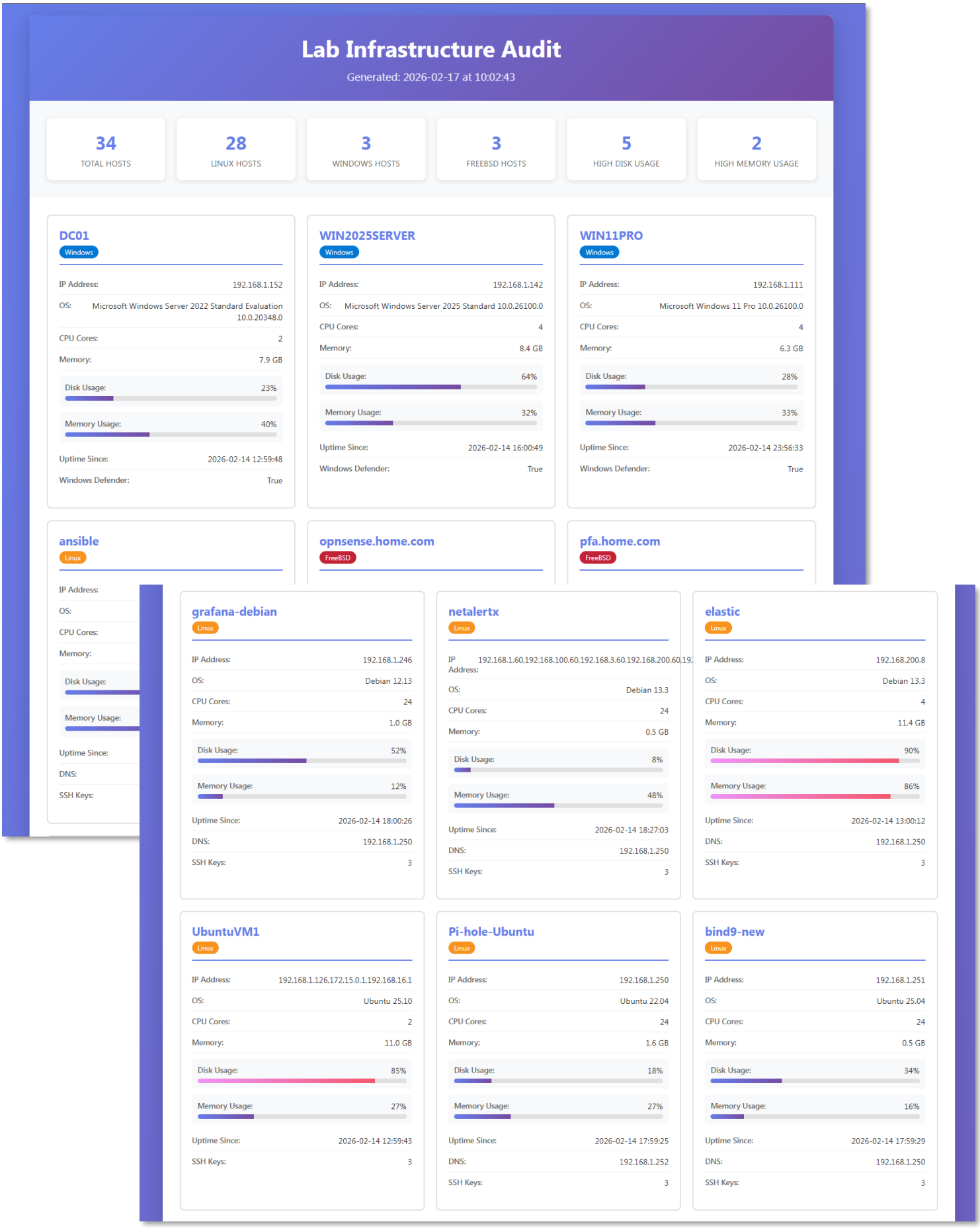
HTML Report Features

- Responsive grid layout with host cards
- Platform-specific color coding (Linux/Windows/FreeBSD)
- Visual metrics with progress bars for disk/memory usage
- Warning/critical thresholds highlighted (>75% yellow, >90% red)
- Summary statistics cards (total hosts, high disk/memory usage counts)
- Timestamp and audit metadata in footer

Apache webserver landing page:



Example Audit Report:



Example Email and Discord Alerts:

n8n Automated System Audit Alert

Delete | Archive | Report | Move to | Reply | Zoom | Quick steps | Read / Unread | Categorize | Flag / Unflag | Print

n8n Automated System Audit Alert

pnleone17@gmail.com
To: soc@shadowitlab.com

Security Operations Center Alert

From: n8n Automated System Audits

To: Security Team

Subject: audit_2026-02-17T12-04-26.html uploaded to Apache webserver

---

Summary

Lab Infrastructure Audit Workflow Completed, Apache webserver updated, audit_2026-02-17T12-04-26.html uploaded to the internal Web Portal.

Details

Timestamp: 2026-02-17 @ 10:02:43

Total Hosts: 34

Summary by OS Family

- **Windows**: 3 hosts
- **Debian**: 24 hosts
- **FreeBSD**: 3 hosts
- **RedHat**: 4 hosts

Summary by OS Distribution

- **Microsoft Windows Server 2022 Standard Evaluation 10.0.203**
- **Microsoft Windows Server 2025 Standard 10.0.26100.0**: 1 host
- **Microsoft Windows 11 Pro 10.0.26100.0**: 1 host
- **Debian 18.0-bookworm-amd64**: 1 host
- **FreeBSD 14.3**: 1 host
- **FreeBSD 15.0**: 2 hosts
- **Debian 12.13**: 10 hosts
- **Debian 13.3**: 5 hosts
- **Ubuntu 25.10**: 3 hosts
- **Ubuntu 22.04**: 3 hosts
- **Ubuntu 25.04**: 2 hosts
- **RedHat 10.1**: 2 hosts
- **Fedora 43**: 1 host
- **CentOS 9**: 1 host

-- End of report --

-- End of message --

This email was sent automatically with n8n

ansible APP 7:39 AM

Lab Infrastructure Audit Workflow Completed

Apache webserver updated, audit_2026-02-17T12-04-26.html uploaded to the internal portal. Report completed 2026-02-17.

Total Hosts: 34

Summary by OS Family

- **Windows**: 3 hosts
- **Debian**: 24 hosts
- **FreeBSD**: 3 hosts
- **RedHat**: 4 hosts

Summary by OS Distribution

- **Microsoft Windows Server 2022 Standard Evaluation 10.0.20348.0**: 1 host
- **Microsoft Windows Server 2025 Standard 10.0.26100.0**: 1 host
- **Microsoft Windows 11 Pro 10.0.26100.0**: 1 host
- **Debian 18.0-bookworm-amd64**: 1 host
- **FreeBSD 14.3**: 1 host
- **FreeBSD 15.0**: 2 hosts
- **Debian 12.13**: 10 hosts
- **Debian 13.3**: 5 hosts
- **Ubuntu 25.10**: 3 hosts
- **Ubuntu 22.04**: 3 hosts
- **Ubuntu 25.04**: 2 hosts
- **RedHat 10.1**: 2 hosts
- **Fedora 43**: 1 host
- **CentOS 9**: 1 host

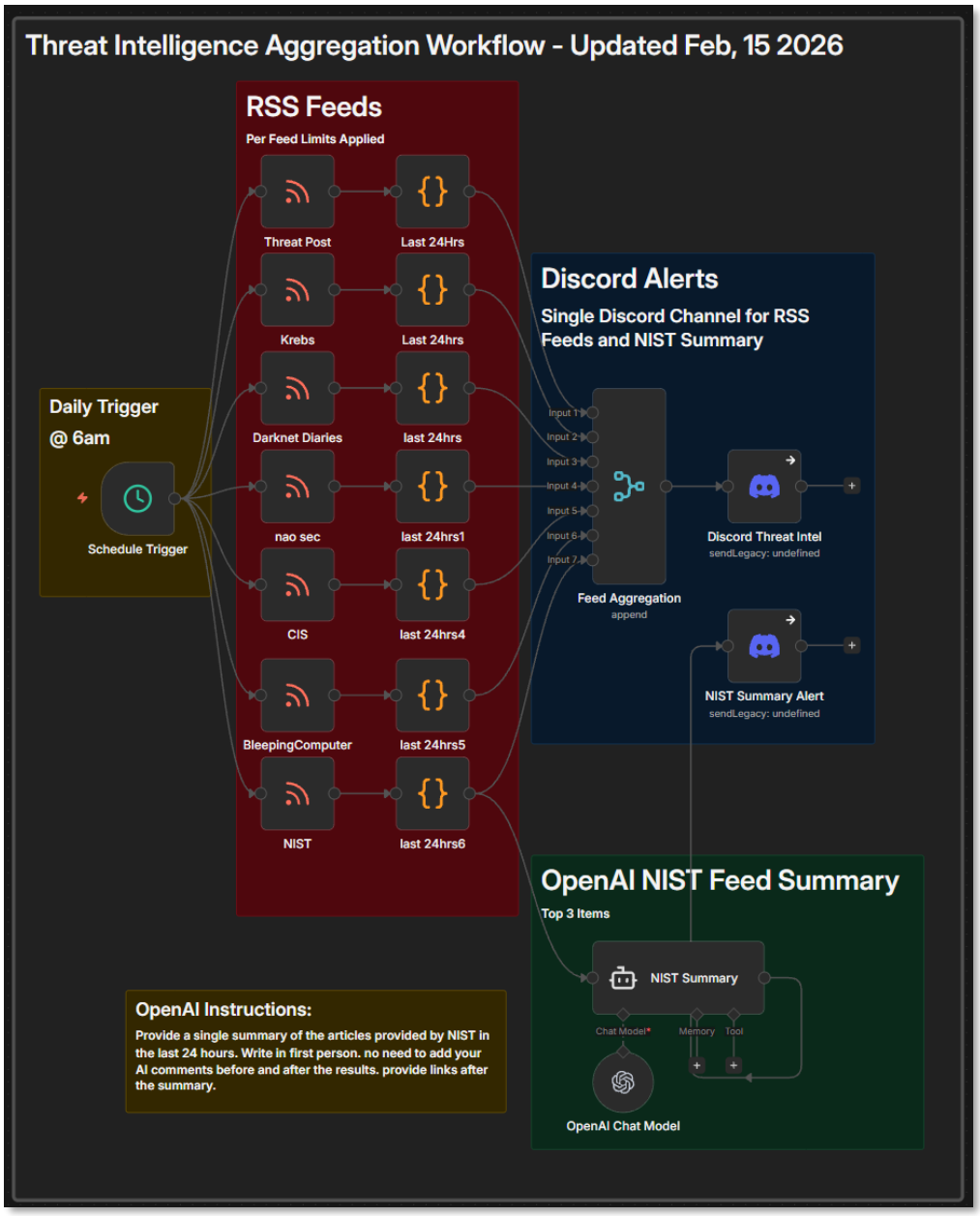
Workflow 2: Threat Intelligence Aggregation

Purpose: Daily ingestion and distribution of curated cybersecurity threat intelligence with AI-powered summarization

This workflow runs daily at 8am to ingest and distribute curated threat intelligence from multiple cybersecurity RSS feeds. It supports situational awareness and IOC enrichment across the lab environment.

Workflow Summary

- Scheduled Execution: Triggered daily via n8n's Cron node (6 AM)
- RSS Feed Polling: Pulls entries from curated list of cybersecurity sources
- Feed Limiting: Filters each feed to articles added in the last 24 hours to reduce noise
- ChatGPT Integration: NIST feed summarized via OpenAI API
- Discord Notification: Formatted aggregated feed summary to #threat-intel channel



RSS Feeds

- Darknet Diaries (https://podcast.darknetdiaries.com/)
- NIST (https://www.nist.gov/blogs/cybersecurity-insights/rss.xml)
- Krebs on Security (https://krebsonsecurity.com/feed/)
- Threat Post (https://threatpost.com/feed/)
- BleepingComputer (https://www.bleepingcomputer.com/feed/)
- CIS (https://www.cisecurity.org/feed/advisories)
- NAO SEC (https://nao-sec.org/feed)

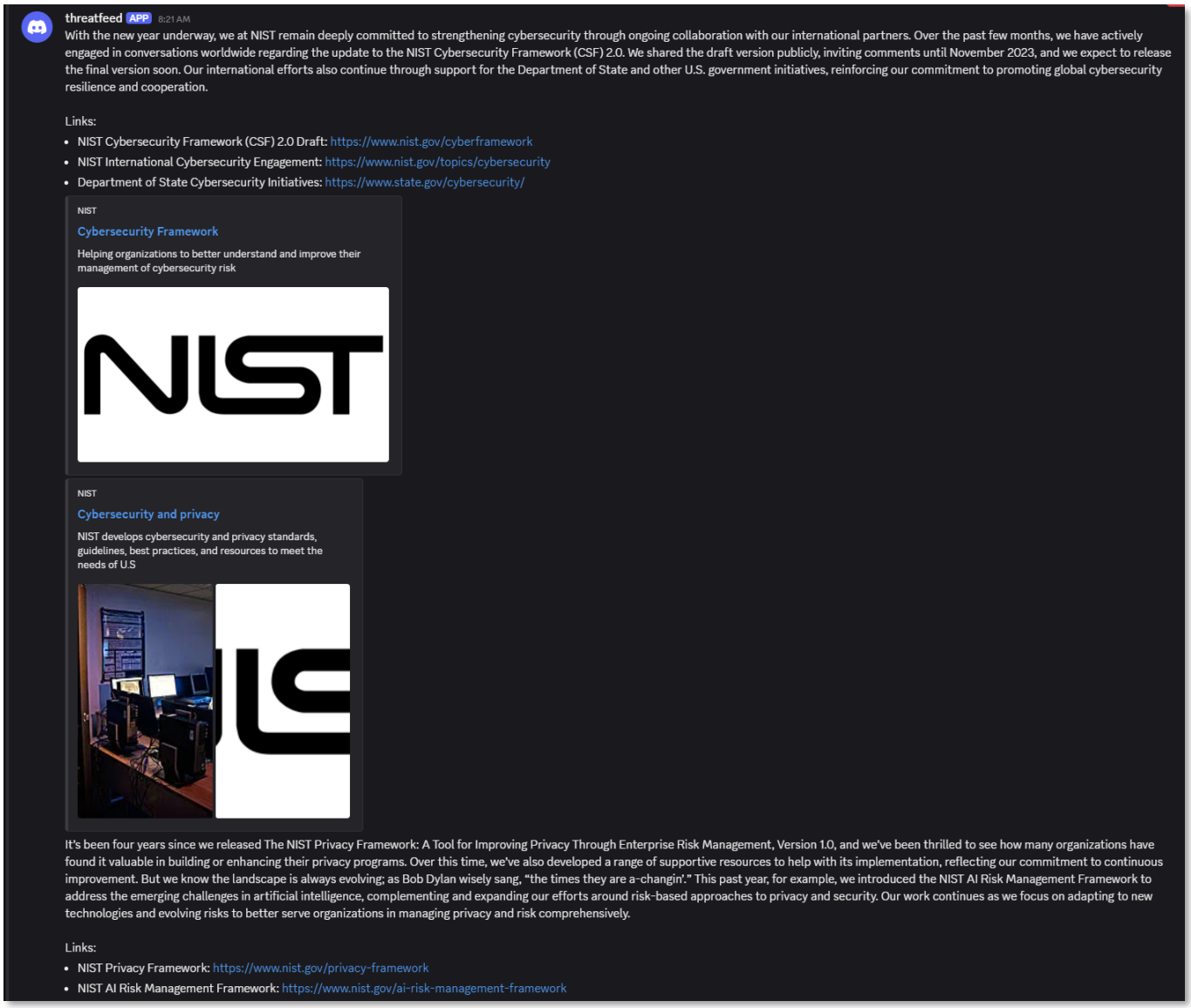
Workflow Nodes

Node Type	Configuration	Purpose
Schedule Trigger	Cron: 0 8 * * * (Daily 6 AM)	Daily execution
RSS Feed Reader	URLs: CIS, NIST, Krebs, ThreatPost, BleepingComputer, etc	Ingest threat intel
Filter Node	Limit each feed to the last 24 hours	Reduce noise
Merge Node	Combine all feeds	Aggregate data
OpenAI Node	Summarize NIST feed with ChatGPT	AI-powered summary
Format Node	Create Discord embed message	Visual formatting
Discord Webhook	POST to #threat-intel channel	Distribute to team

Workflow Benefits

- Centralized threat intelligence (7 sources → 1 channel)
- AI-powered summarization reduces information overload
- Daily cadence ensures timely awareness of emerging threats
- Supports incident response and vulnerability management

NIST Summary Alert:



Example Discord Alerts:

threatfeed APP 12:47 PM
Tue, 17 Feb 2026 10:37:45 -0500
Microsoft Teams outage affects users in United States, Europe
https://www.bleepingcomputer.com/news/microsoft/microsoft-tean
___
Creator: Sergiu Gatlan

BleepingComputer
Microsoft Teams outage affects users in United States, Europe
Microsoft is working to resolve an ongoing outage affecting Microsoft Teams users, causing delays and preventing some from accessing the service.

Tue, 17 Feb 2026 09:40:49 -0500
What 5 Million Apps Revealed About Secrets in JavaScript
https://www.bleepingcomputer.com/news/security/what-5-million-a
___
Creator: Sponsored by Intruder

BleepingComputer
What 5 Million Apps Revealed About Secrets in JavaScript
Leaked API keys are nothing new, but the scale of the problem in front-end code has been largely a mystery - until now. Intruder's research team built a new secrets detection method and scanned 5 million applications specifically looking for secrets hidden in JavaScript bundles. Here's what we learned.

Tue, 17 Feb 2026 09:05:25 -0500
New Keenadu backdoor found in Android firmware, Google Play apps
https://www.bleepingcomputer.com/news/security/new-keenadu-ba

The Washington Hotel brand in Japan has announced that that its servers were compromised in a ransomware attack, exposing various business data.

Mon, 16 Feb 2026 14:19:09 -0500
Eurail says stolen traveler data now up for sale on dark web
https://www.bleepingcomputer.com/news/security/eurail-says-stolen-traveler-data-now-u
___
Creator: Bill Toulas

BleepingComputer
Eurail says stolen traveler data now up for sale on dark web
Eurail B.V., the operator that provides access to 250,000 kilometers of European railways, confirmed that data stolen in a breach earlier this year is being offered for sale on the dark web.

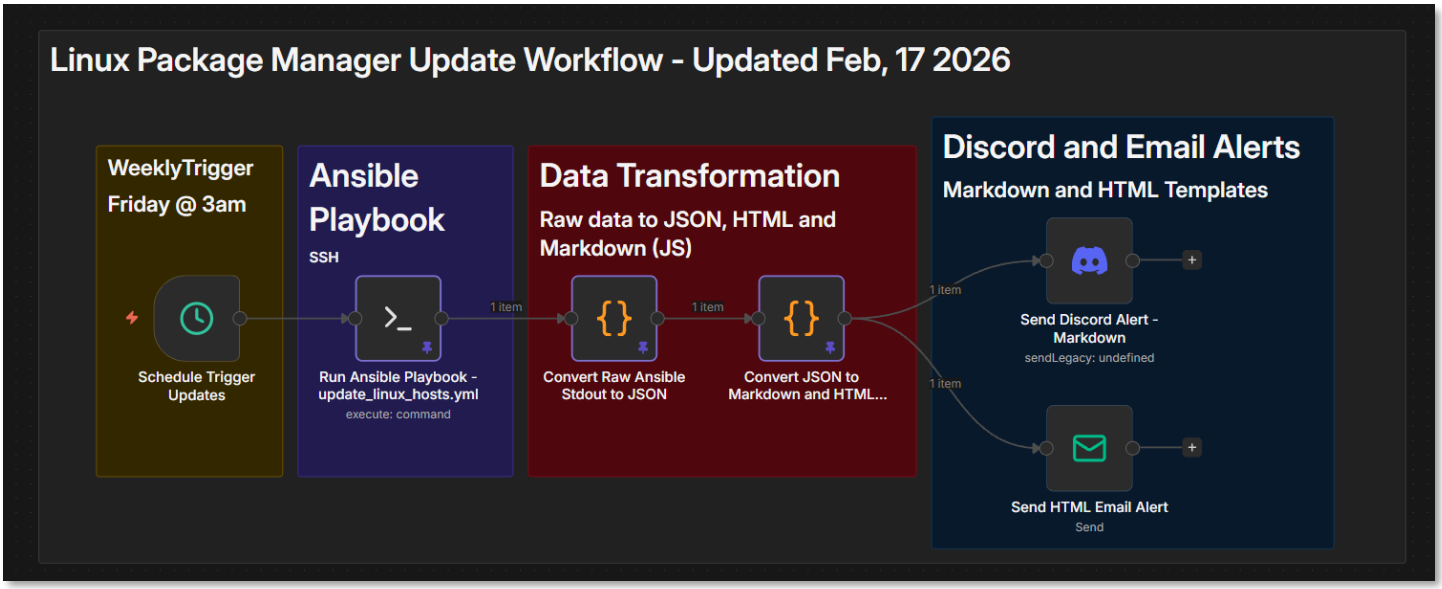
Mon, 16 Feb 2026 14:13:39 -0500
Man arrested for demanding reward after accidental police data leak
https://www.bleepingcomputer.com/news/security/man-arrested-for-demanding-reward-a
___
Creator: Sergiu Gatlan

BleepingComputer
Man arrested for demanding reward after accidental police data leak
Dutch authorities arrested a 40-year-old man after he downloaded confidential documents that had been mistakenly shared by the police and refused to delete them unless he received "something in return."

Workflow 3: Weekly Package Manager Updates with Alerting

Purpose: Automated weekly package manager updates across all Linux hosts with Discord and email alerting

This workflow runs weekly on Friday at 3am, executes the update_linux_hosts.yml Ansible playbook, parses the output to categorize update results, and sends formatted notifications via Discord and email.



Workflow Summary

- Scheduled Execution: Weekly trigger (Friday 3 AM)
- Ansible Playbook: Executes update_linux_hosts.yml for apt/dnf updates
- Data Parsing: Converts raw Ansible stdout to structured JSON
- Update Categorization: Hosts grouped by status (no updates, updated, reboot required, errors)
- Markdown Generation: Creates formatted summary for Discord
- HTML Generation: Creates formatted report for email
- Discord Alert: Sends markdown summary with status counts
- Email Alert: Sends HTML email with detailed update report

Workflow Nodes

Node Type	Configuration	Purpose
Schedule Trigger	Cron: 0 3 * * 5 (Friday 3 AM)	Weekly execution
SSH Node	ansible-playbook update_linux_hosts.yml	Run package updates
Code Node 1	Parse stdout: extract reachable/unreachable hosts, summaries, recap	Convert to structured JSON
Code Node 2	Generate markdown and HTML summaries with categorization	Format alert content
Discord Node	POST markdown summary	Send Discord notification
Email Node	SMTP send HTML report	Send email notification

Data Flow and Parsing Logic

The workflow parses Ansible's stdout output to extract comprehensive update information:

- Reachable Hosts: Extracted from 'ok: [hostname]' lines
- Unreachable Hosts: Extracted from 'fatal: [hostname]' lines
- Update Summaries: Parsed from 'msg' fields containing platform, packages updated, reboot status
- Play Recap: Structured data showing ok/changed/failed/skipped counts per host

Host Categorization

Hosts are automatically categorized based on update results:

- No Updates Required: Hosts with packages_updated = False
- Updates Applied: Hosts with packages_updated = True and reboot_required = False
- Reboot Required: Hosts with reboot_required = True
- Errors Detected: Hosts with empty/null platform, packages_updated, reboot_required
- Unreachable: Hosts that failed connectivity checks

Code Snippet - Stdout to JSON Parsing

```
// Parse Ansible stdout into structured data
const stdout = $input.first().json.stdout;
const lines = stdout.split('\n');

// Extract reachable and unreachable hosts
let reachable = [];
let unreachable = [];

for (const line of lines) {
  const okMatch = line.match(/^ok:\s+\[(.*?)\]/);
  if (okMatch) reachable.push(okMatch[1]);

  const fatalMatch = line.match(/^fatal:\s+\[(.*?)\]/);
  if (fatalMatch) unreachable.push(fatalMatch[1]);
}

// Parse update summaries from msg fields
let summaries = [];
let currentHost = null;

for (const line of lines) {
  const hostMatch = line.match(/^ok:\s+\[(.*?)\]\s+=>/);
  if (hostMatch) currentHost = hostMatch[1];

  if (line.includes('"msg":')) {
    const msg = line.replace(/.*"msg":\s+"|",?$/g, "");
    const platform = (msg.match(/Platform:\s+(.*)/) || [])[1];
    const updated = (msg.match(/Packages updated:\s+(.*)/) || [])[1];
    const reboot = (msg.match(/Reboot required:\s+(.*)/) || [])[1];

    summaries.push({
      host: currentHost,
      platform,
      packages_updated: updated,
      reboot_required: reboot
    });
  }
}

return [{ json: { total_hosts, reachable_hosts, unreachable_hosts,
update_summaries, recap } }];
```

Code Snippet - Categorization and Formatting

```
// Categorize hosts by update status
const noUpdates = [];
const updatesApplied = [];
const rebootRequired = [];
const errorsDetected = [];

for (const s of normalized) {
  if (s.error) {
    errorsDetected.push(s.host);
  } else if (s.reboot) {
    rebootRequired.push(s.host);
  } else if (s.updated) {
    updatesApplied.push(s.host);
  } else {
    noUpdates.push(s.host);
  }
}

// Generate markdown for Discord
const md = `
## Linux Update Summary

**Total Hosts:** ${total}
**Reachable:** ${reachable.length}
**Unreachable:** ${unreachable.length}

🟢 No Updates Required (${noUpdates.length})
${noUpdates.map(h => \`- ${h}\`).join("\n")}

🔵 Updates Applied (${updatesApplied.length})
${updatesApplied.map(h => \`- ${h}\`).join("\n")}

🟠 Reboot Required (${rebootRequired.length})
${rebootRequired.map(h => \`- ${h}\`).join("\n")}

🔴 Errors Detected (${errorsDetected.length})
${errorsDetected.map(h => \`- ${h}\`).join("\n")}
`;

return [{ json: { markdown: md, html: html } }];
```

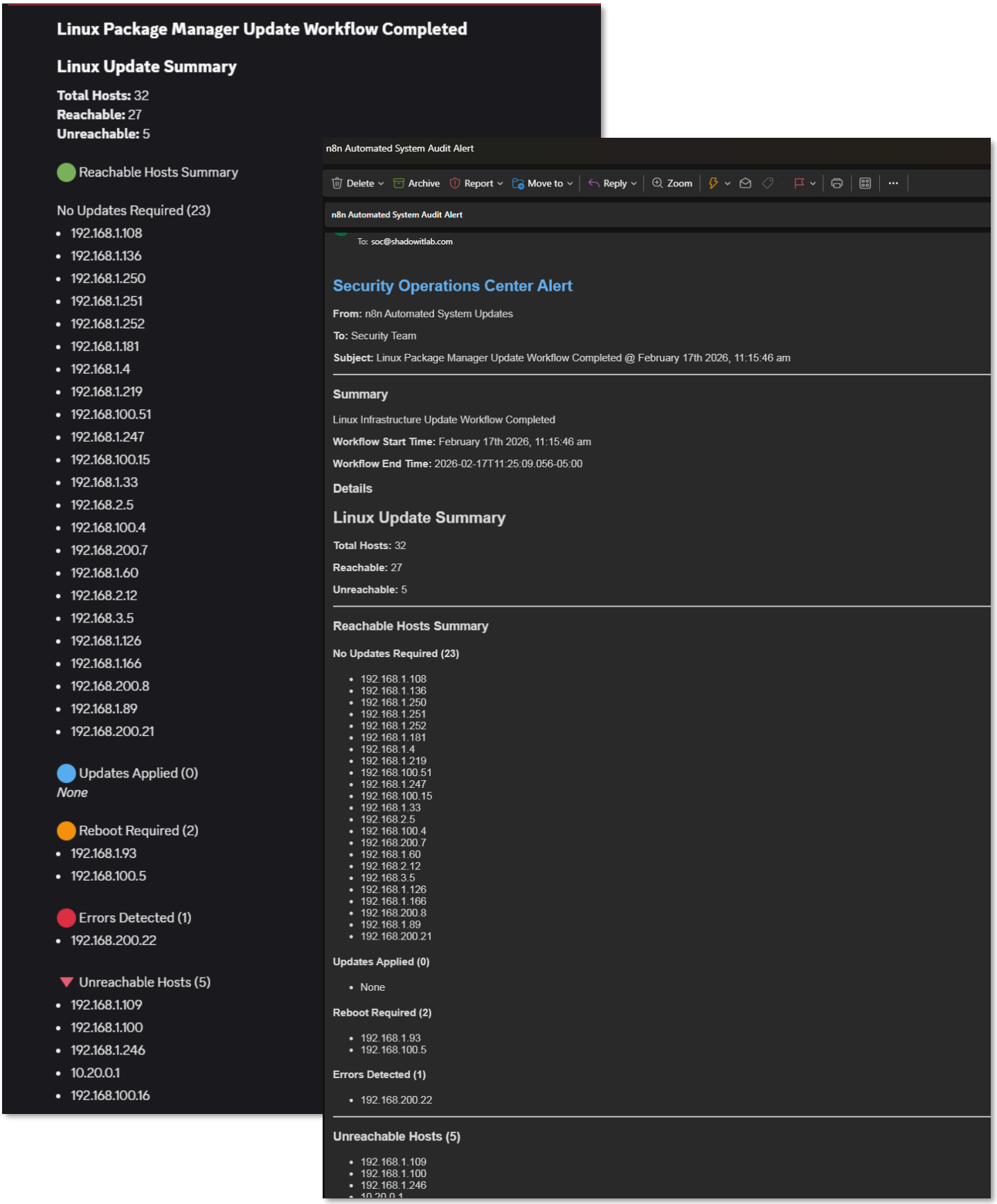
Workflow Benefits

- Automated weekly package updates reduce manual maintenance
- Categorized results prioritize attention (errors and reboots first)
- Dual alerting ensures visibility across communication channels
- Structured parsing enables trend analysis over time
- Unreachable host detection prevents silent failures

Use Cases

- Weekly security patch deployment
- Compliance maintenance (systems up-to-date)
- Post-vulnerability scanning remediation
- Reboot planning based on kernel updates
- Infrastructure health monitoring

Discord and Email Alert Example:



Linux Package Manager Update Workflow Completed

Linux Update Summary

Total Hosts: 32
Reachable: 27
Unreachable: 5

Reachable Hosts Summary

No Updates Required (23)

- 192.168.1.108
- 192.168.1.136
- 192.168.1.250
- 192.168.1.251
- 192.168.1.252
- 192.168.1.181
- 192.168.1.4
- 192.168.1.219
- 192.168.100.51
- 192.168.1.247
- 192.168.100.15
- 192.168.1.33
- 192.168.2.5
- 192.168.100.4
- 192.168.200.7
- 192.168.1.60
- 192.168.2.12
- 192.168.3.5
- 192.168.1.126
- 192.168.1.166
- 192.168.200.8
- 192.168.1.89
- 192.168.200.21

Updates Applied (0)
None

Reboot Required (2)

- 192.168.1.93
- 192.168.100.5

Errors Detected (1)

- 192.168.200.22

Unreachable Hosts (5)

- 192.168.1.109
- 192.168.1.100
- 192.168.1.246
- 10.20.0.1
- 192.168.100.16

n8n Automated System Audit Alert

Delete | Archive | Report | Move to | Reply | Zoom

n8n Automated System Audit Alert

To: soc@shadowitlab.com

Security Operations Center Alert

From: n8n Automated System Updates

To: Security Team

Subject: Linux Package Manager Update Workflow Completed @ February 17th 2026, 11:15:46 am

Summary

Linux Infrastructure Update Workflow Completed

Workflow Start Time: February 17th 2026, 11:15:46 am

Workflow End Time: 2026-02-17T11:25:09.056-05:00

Details

Linux Update Summary

Total Hosts: 32

Reachable: 27

Unreachable: 5

Reachable Hosts Summary

No Updates Required (23)

- 192.168.1.108
- 192.168.1.136
- 192.168.1.250
- 192.168.1.251
- 192.168.1.252
- 192.168.1.181
- 192.168.1.4
- 192.168.1.219
- 192.168.100.51
- 192.168.1.247
- 192.168.100.15
- 192.168.1.33
- 192.168.2.5
- 192.168.100.4
- 192.168.200.7
- 192.168.1.60
- 192.168.2.12
- 192.168.3.5
- 192.168.1.126
- 192.168.1.166
- 192.168.200.8
- 192.168.1.89
- 192.168.200.21

Updates Applied (0)

- None

Reboot Required (2)

- 192.168.1.93
- 192.168.100.5

Errors Detected (1)

- 192.168.200.22

Unreachable Hosts (5)

- 192.168.1.109
- 192.168.1.100
- 192.168.1.246
- 10.20.0.1

Workflow 4: Monthly Automated Ansible Token Rotation

[PLACEHOLDER - Implementation Pending]

Purpose: Automated monthly credential rotation for Ansible vault with random token generation, vault file update, user management playbook execution, and dual alerting

Planned Workflow Summary

- Scheduled Execution: Monthly trigger (1st of month at 4 AM)
- Token Generation: Create cryptographically random 32-char token (base64)
- Vault Update: SSH to Ansible controller and update vault_ansible_password in vault.yml
- Playbook Execution: Run user_mgmt.yml --tags ansible_user to propagate new password
- Connectivity Test: Verify Ansible can still connect to all hosts via ping
- Discord Alert: Send success/failure notification with rotation summary
- Email Alert: Send HTML email confirming rotation and test results

Planned Workflow Nodes

Node Type	Configuration	Purpose
Schedule Trigger	Cron: 0 4 1 * * (1st of month 4 AM)	Monthly execution
Code Node 1	Generate token: crypto.randomBytes(24).toString('base64')	Create new password
SSH Node 1	Backup current vault.yml	Create vault backup
SSH Node 2	ansible-vault edit vault.yml (update vault_ansible_password)	Update vault variable
SSH Node 3	ansible-playbook user_mgmt.yml --tags ansible_user	Propagate new password
SSH Node 4	ansible linux -m ping	Test connectivity
Code Node 2	Parse ping results for success count	Verify all hosts accessible
IF Node	Check if ping success == total hosts	Route to success/failure
Code Node 3	Generate Discord success message	Format success alert
Discord Webhook	POST rotation success to Discord	Send Discord notification
Code Node 4	Generate HTML email template	Format email report
Email Node	SMTP send with rotation details	Send email notification
Error Handler	Rollback vault.yml and alert on failure	Handle rotation errors

Security Considerations for Token Rotation

- Token generation uses crypto.randomBytes (cryptographically secure)
- Vault backup created before each rotation for rollback capability
- Connectivity test verifies all hosts accessible before confirming rotation
- Error handler automatically reverts to backup vault on failure
- Credentials never logged or displayed in workflow execution history
- n8n credential vault stores vault password with AES-256 encryption

Expected Alert Content

- Rotation timestamp
- Token generation success
- Vault update status
- User management playbook execution result
- Connectivity test results (hosts reachable/unreachable)
- Rollback status (if applicable)

Best Practices and Lessons Learned

Error Handling and Resilience

- All SSH nodes include timeout settings (30-60 seconds)
- Workflow execution logs retained for 30 days in n8n database

Credential Management

- All credentials stored in n8n vault (never hardcoded)
- SSH credentials use key-based auth where possible
- Webhook URLs stored as environment variables
- SMTP credentials encrypted with AES-256 at rest

Notification Design

- Discord: Concise markdown with key metrics and report links
- Email: Detailed HTML templates with embedded CSS for compatibility
- Critical alerts include @ mentions for immediate attention
- All notifications include timestamp and workflow execution ID

Integration Patterns

- Ansible workflows use SSH node for remote execution
- JSON output from playbooks parsed in Code nodes
- HTML and Markdown generation via JavaScript templates
- Apache webserver deployment via SSH file write operations
- Multi-platform support handled through conditional Ansible blocks

Scripting for Advanced Automation

Script Development Strategy

Custom scripts supplement configuration management tools for tasks requiring complex logic, performance optimization, or specialized functionality. All scripts are version-controlled in Git, documented with inline comments, and integrated into broader automation workflows.

Security Impact

- Complex security operations automated where configuration-management tools lack native functionality
- Performance-critical tasks (log parsing, threat-intelligence queries) optimized beyond built-in tool capabilities
- Security tool APIs integrated through custom automation wrappers for expanded functionality
- Rapid prototyping enables validation of security concepts before production-grade implementation

Deployment Rationale: While Ansible/Terraform handle declarative configuration, procedural logic (API rate-limiting, stateful workflows, real-time processing) requires traditional scripting. Enterprise environments use scripts for custom integrations, API gateways, and performance-critical operations. This demonstrates ability to select appropriate automation tools—declarative vs. imperative—based on task requirements.

Architecture Principles Alignment

- **Defense in Depth:** Scripts validate input parameters; error handling prevents cascading failures; execution logs audited for anomalies
- **Secure by Design:** No hardcoded credentials (environment variables or secret managers); input sanitization prevents injection attacks; least-privilege execution (non-root where possible)
- **Zero Trust:** Every script execution logged with user/timestamp; API calls authenticated via short-lived tokens; output validation before downstream processing

Script Language Selection Criteria:

Language	Use Cases	Advantages
Bash	Linux system administration; cron jobs	Native; fast; no dependencies
PowerShell	Windows management; AD operations	Deep Windows integration; objects
Python	API integration; data processing; ML	Rich libraries; cross-platform

PowerShell & Bash Scripting

Custom automation scripts are developed in both PowerShell (for Windows systems) and Bash (for Linux systems) to handle repetitive administrative tasks, enforce configuration standards, and respond to system events. These scripts automate activities such as user account provisioning, log rotation, backup verification, certificate renewal checks, and security baseline enforcement. PowerShell scripts leverage native Windows management frameworks like Active Directory modules and WMI, while Bash scripts utilize standard Unix utilities and interact with system APIs. Scripts are version-controlled in GitHub alongside infrastructure code, enabling rollback capability and documentation of automation logic.

Bash Script Example: Backup Automation

This Bash script is designed to perform a backup using rsync, with versioned backups stored by date. It validates input, checks for dependencies, and logs the simulated operation.


```
Bash > $ backup.sh > ...
#!/bin/bash

# check to make sure the user has entrered exactly two arguments.
if [ $# -ne 2 ]
then
    /usr/bin/echo "Usage: backup.sh <source_directory> <target_directory>"
    /usr/bin/echo "Please try again."
    exit 1
fi

#check to see if rsync is installed
if ! command -v rsync > /dev/null 2>&1
then
    /usr/bin/echo "This script requires rsync to be installed."
    /usr/bin/echo "Please install the package and run the script again."
    exit 2
fi

# Capture the current date, and store it in the format YYYY-MM-DD
current_date=$(date +%Y-%m-%d)

rsync_options="-avb --backup-dir $2/$current_date --delete --dry-run" #tested with dry-run prior to adding into production,

$(which rsync) $rsync_options $1 $2/current >> /var/log/backup_$current_date.log

```

Script: /usr/local/bin/backup_web.sh

Purpose: Incremental rsync backup with versioning and logging

Line(s)	Purpose	Explanation
#!/bin/bash	Script interpreter	Ensures the script runs using Bash.
if [$# -ne 2]	Argument check	Verifies that exactly two arguments are provided.
echo "Usage:..."	Usage message	Informs the user of correct syntax if arguments are missing.

exit 1	Exit on error	Terminates with exit code 1 for incorrect usage.
command -v rsync	Dependency check	Verifies that rsync is installed and available in $PATH.
exit 2	Exit on missing dependency	Terminates with exit code 2 if rsync is not found.
current_date=$(date +%Y-%m-%d)	Timestamp	Captures the current date in YYYY-MM-DD format for versioning.
rsync_options=...	Backup options	Sets rsync flags: • -a: archive mode (preserves permissions, timestamps, etc.) • -v: verbose • -b: backup files before overwriting • --backup-dir: stores backups in a dated subdirectory • --delete: removes files in target that no longer exist in source • --dry-run: simulates the operation without making changes
$(which rsync) ...	Execute rsync	Runs rsync with the defined options, syncing from $1 (source) to $2/current (target).
>> /var/log/backup_$current_date.log	Logging	Appends output to a date-stamped log file for auditability.

Linux Upgrade Script Overview

This Bash script performs a distribution-aware system upgrade, logging all output and errors to dedicated log files. It supports Debian/Ubuntu, Fedora, CentOS, and RHEL, and includes error checking after each upgrade step

```bash
#!/bin/bash

logfile=/var/log/update_script.log
errorlog=/var/log/update_script_errors.log

/usr/bin/echo "------------------START SCRIPT-------------------------" 1>>$logfile 2>>$errorlog
check_exit_status() {
    if [ $? -ne 0 ]
    then
        /usr/bin/echo "An error occured, please check the $errorlog file."
    fi
}

if [ -d /etc/apt ]; then
    # Debian or Ubuntu
    /usr/bin/echo "Detected Debian/Ubuntu system"
    /usr/bin/sudo apt update 1>>$logfile 2>>$errorlog
    check_exit_status
    /usr/bin/sudo apt dist-upgrade -y 1>>$logfile 2>>$errorlog
    check_exit_status
elif [ -f /etc/redhat-release ]; then
    distro=$(cat /etc/redhat-release)

    if [[ "$distro" == *"Fedora"* ]]; then
        /usr/bin/echo "Detected Fedora system"
        /usr/bin/sudo dnf upgrade --refresh -y #!/bin/bash

logfile=/var/log/update_script.log
errorlog=/var/log/update_script_errors.log


check_exit_status() {
    if [ $? -ne 0 ]
    then
        /usr/bin/echo "An error occured, please check the $errorlog file."
    fi
}

if [ -d /etc/apt ]; then
    # Debian or Ubuntu
    /usr/bin/echo "Detected Debian/Ubuntu system"
    /usr/bin/sudo apt update 1>>$logfile 2>>$errorlog
    check_exit_status
    /usr/bin/sudo apt dist-upgrade -y 1>>$logfile 2>>$errorlog
    check_exit_status
elif [ -f /etc/redhat-release ]; then
    distro=$(cat /etc/redhat-release)

    if [[ "$distro" == *"Fedora"* ]]; then
        /usr/bin/echo "Detected Fedora system"
        /usr/bin/sudo dnf upgrade --refresh -y 1>>$logfile 2>>$errorlog
        check_exit_status
    elif [[ "$distro" == *"CentOS"* ]] || [[ "$distro" == *"Red Hat"* ]]; then
        /usr/bin/echo "Detected CentOS or RHEL system"
        /usr/bin/sudo yum update -y 1>>$logfile 2>>$errorlog
        check_exit_status
    else
        /usr/bin/echo "Detected unknown Red Hat-based system"
        /usr/bin/sudo yum update -y 1>>$logfile 2>>$errorlog
        check_exit_status
    fi
else
    /usr/bin/echo "Unsupported or unknown Linux distribution"
fi

/usr/bin/echo "The script completed at: $(/usr/bin/date)" 1>>$logfile 2>>$errorlog
/usr/bin/echo "------------------END SCRIPT-------------------------" 1>>$logfile 2>>$er
```

Line(s)	Purpose	Explanation
#!/bin/bash	Interpreter declaration	Ensures the script runs with Bash.

logfile=/var/log/update_script.log **errorlog=/var/log/update_script_errors.log**	Log file setup	Defines paths for standard output and error logs.
echo "START SCRIPT"	Start marker	Logs the beginning of the script execution.
check_exit_status()	Error handler	Function that checks the last command's exit code and prints an error message if non-zero.
if [-d /etc/apt]; then	Distro detection	Checks for Debian/Ubuntu by presence of APT directory.
apt update **apt dist-upgrade -y**	Debian/Ubuntu upgrade	Runs update and full upgrade, logs output and errors.
elif [-f /etc/redhat-release]; then	Red Hat-based detection	Checks for Fedora, CentOS, or RHEL using release file.
cat /etc/redhat-release	Distro name	Reads the release file to identify the specific variant.
dnf upgrade --refresh -y	Fedora upgrade	Refreshes metadata and upgrades packages.
yum update -y	CentOS/RHEL upgrade	Performs system update using yum.
echo "Unsupported distro"	Fallback	Handles unknown or unsupported systems.
echo "Script completed at: $(date)"	Completion timestamp	Logs the end time of the script.
echo "END SCRIPT"	End marker	Logs the conclusion of the script execution.

PowerShell Script Example: Windows Update Automation

This script performs a comprehensive update sweep across a Windows system, covering:

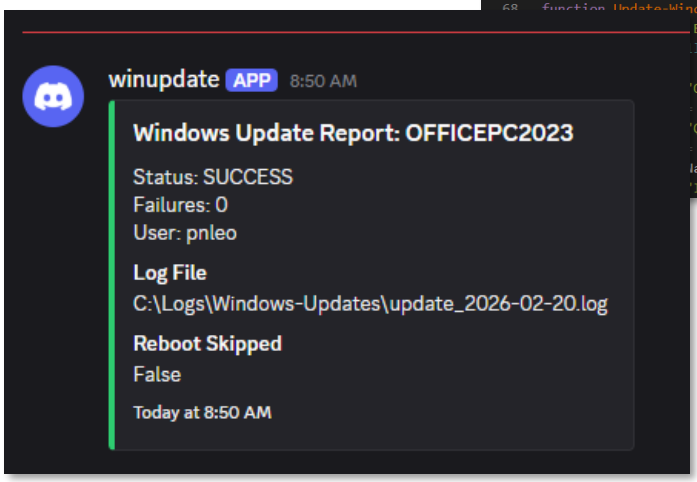
- Windows Store apps
- Chocolatey packages
- Winget packages
- Windows OS updates

All output is captured to a transcript file with timestamps for auditability and runtime tracking. The script handles missing tools gracefully (skip vs fail), fixes the EventLog 32,766-char limit, silently pre-installs NuGet, and sends a Discord webhook summary on completion.

Script: C:\Scripts\Update-System.ps1

Purpose: Comprehensive Windows update across multiple package managers

```powershell
#>

[CmdletBinding()]
param(
    [switch]$SkipReboot,
    [string]$WebhookUrl = "https://discord.com/api/webhooks/1435986566647644201/S1Lb6UFEBs_cLzGK9dl70WZpifZpbT
)

$ErrorActionPreference = "Continue"
$LogPath               = "C:\Logs\Windows-Updates"
$LogFile               = Join-Path $LogPath "update_$(Get-Date -Format 'yyyy-MM-dd').log"
$MaxEventLogMsgLength  = 30000
$FailureCount          = 0

if (-not (Test-Path $LogPath)) {
    New-Item -ItemType Directory -Path $LogPath -Force | Out-Null
    Write-Host "[BOOTSTRAP] Created log directory: $LogPath"
} else {
    Write-Host "[BOOTSTRAP] Log directory exists: $LogPath"
}

Start-Transcript -Path $LogFile -Append
Write-Host "[BOOTSTRAP] Transcript started: $LogFile"

Write-Host "[BOOTSTRAP] Checking NuGet provider..."
$nuget = Get-PackageProvider -Name NuGet -ErrorAction SilentlyContinue
if (-not $nuget -or $nuget.Version -lt [Version]"2.8.5.201") {
    Write-Host "[BOOTSTRAP] Installing NuGet provider..."
    Install-PackageProvider -Name NuGet -MinimumVersion 2.8.5.201 -Force -Scope AllUsers | Out-Null
    Write-Host "[BOOTSTRAP] NuGet provider installed."
} else {
    Write-Host "[BOOTSTRAP] NuGet provider OK (version $($nuget.Version))."
}

74 references
function Write-Log {
    param(
        [string]$Message,
        [string]$Level = "INFO"
    )
    $Timestamp = Get-Date -Format "yyyy-MM-dd HH:mm:ss"
    Write-Output "[$Timestamp] [$Level] $Message"

    $EventMessage = if ($Message.Length -gt $MaxEventLogMsgLength) {
        $Message.Substring(0, $MaxEventLogMsgLength) + "`n[TRUNCATED - see log file]"
    } else {
        $Message
    }
    $EventId = switch ($Level) { "ERROR" { 3 } "WARNING" { 2 } default { 1 } }
    Write-EventLog -LogName Application -Source "WindowsUpdate" `
        -EntryType Information -EventId $EventId -Message $EventMessage `
        -ErrorAction SilentlyContinue
}

3 references
function Update-WindowsStoreApps {
    BEGIN: Windows Store App Updates ---"
    l

    Opening CIM session..."
     New-CimSession
    Querying MDM AppManagement class..."
     Get-CimInstance -Namespace "root\cimv2\mdm\dmmap" `
    lame "MDM_EnterpriseModernAppManagement_AppManagement01"
    Invoking Store update scan..."
```



Line / Block	Purpose	Explanation
$ErrorActionPreference = "Continue"	Error handling	Script continues if a command fails. Individual function returns false to increment $FailureCount.
$LogFile = Join-Path ...	Log file path	Defines daily-named log file under C:\Logs\Windows-Updates\.

Start-Transcript / Stop-Transcript	Full session capture	Records all console output to the log file including command output and errors.
function Write-Log { ... }	Timestamped logging	Prefixes every message with timestamp and level tag. Truncates to 30,000 chars before writing to EventLog (hard limit: 32,766).
Get-PackageProvider (NuGet)	NuGet bootstrap	Pre-installs NuGet silently before PSWindowsUpdate install. Prevents interactive Y/N prompt that blocks automated runs.
function Update-WindowsStoreApps { }	Store updates	Uses CIM to call UpdateScanMethod on the MDM_EnterpriseModernAppManagement class. Triggers background Store refresh.
function Update-ChocolateyPackages { }	Chocolatey updates	Checks for choco in PATH. If found, runs 'choco upgrade all -y' and logs full output. Returns true on exit 0.
function Update-WingetPackages { }	Winget updates	Checks for winget in PATH. Runs upgrade --all with agreements accepted. Non-zero exit logs WARNING but does not fail.
function Update-WindowsOS { }	OS patch management	Installs PSWindowsUpdate if missing. Lists each KB/title/size before installing. Respects -SkipReboot flag.
function Send-DiscordNotification { }	Webhook alert	Builds Discord embed payload. Uses ConvertTo-Json -Depth 10 -Compress to fix nested hashtable serialization (Discord error 50109).
if ($FailureCount -eq 0) { exit 0 }	Exit codes	0 = all passed. 1 = 1-2 failures (partial). 2 = 3+ failures (mostly failed). Used by schedulers to detect failure.

Code and Output Examples

Write-Log - Timestamped Logging with EventLog Truncation

```
function Write-Log {
    param([string]$Message, [string]$Level = "INFO")

    $Timestamp = Get-Date -Format "yyyy-MM-dd HH:mm:ss"
    Write-Output "[$Timestamp] [$Level] $Message"

    # Truncate to 30,000 chars - EventLog hard limit is 32,766
    $EventMsg = if ($Message.Length -gt $MaxEventLogMsgLength) {
        $Message.Substring(0, $MaxEventLogMsgLength) + "`n[TRUNCATED - see log
file]"
    } else { $Message }

    $EventId = switch ($Level) { "ERROR" { 3 } "WARNING" { 2 } default { 1 } }
    Write-EventLog -LogName Application -Source "WindowsUpdate" \
        -EntryType Information -EventId $EventId -Message $EventMsg \
        -ErrorAction SilentlyContinue
}
```

Sample Output:

```
[2026-02-20 07:33:08] [INFO]   ========================================
[2026-02-20 07:33:08] [INFO]   ===== Windows Update Script Started =====
[2026-02-20 07:33:08] [INFO]   Computer  : OFFICEPC2023
[2026-02-20 07:33:08] [INFO]   User      : pnleo
[2026-02-20 07:33:08] [INFO]   OS        : Windows 11 Pro
[2026-02-20 07:33:08] [INFO]   PS Version: 5.1.26100.7705
```

Update-WindowsOS - PSWindowsUpdate Module

```
function Update-WindowsOS {
    Write-Log "--- BEGIN: Windows OS Updates ---"
    if (-not (Get-Module -ListAvailable -Name PSWindowsUpdate)) {
        Write-Log "PSWindowsUpdate not found - installing..."
        Install-Module -Name PSWindowsUpdate -Force -Scope AllUsers -Confirm:$false
    }
    try {
        Import-Module PSWindowsUpdate -ErrorAction Stop
        Write-Log "Checking for available updates..."
        $Updates = Get-WindowsUpdate -MicrosoftUpdate -AcceptAll
        if ($Updates -and $Updates.Count -gt 0) {
            Write-Log "Found $($Updates.Count) update(s):"
            foreach ($u in $Updates) {
                Write-Log "  KB$($u.KBArticleIDs)  |  $($u.Title)  |
$([math]::Round($u.Size/1MB,2)) MB"
            }
            Install-WindowsUpdate -MicrosoftUpdate -AcceptAll -AutoReboot:(-not
$SkipReboot)
            Write-Log "--- END: Windows OS Updates [SUCCESS] ---"
        } else {
            Write-Log "No updates available - system is current."
        }
        return $true
    }
    catch {
        Write-Log $_.Exception.Message -Level "ERROR"
        return $false
    }
}
```

Sample Output:

```
[2026-02-20 07:34:56] [INFO]  Step 4 of 4: Windows OS Updates
[2026-02-20 07:34:56] [INFO]  --- BEGIN: Windows OS Updates ---
[2026-02-20 07:34:56] [INFO]  PSWindowsUpdate already installed (version 2.2.1.4).
[2026-02-20 07:34:56] [INFO]  Importing PSWindowsUpdate...
[2026-02-20 07:34:58] [INFO]  Checking for available updates...
[2026-02-20 07:36:10] [INFO]  Found 2 update(s):
[2026-02-20 07:36:10] [INFO]    KB5034441  | 2026-02 Cumulative Update for Windows 11  | 312.45 MB
[2026-02-20 07:36:10] [INFO]    KB890830   | Windows Malicious Software Removal Tool   | 4.12 MB
[2026-02-20 07:36:10] [INFO]  Installing (AutoReboot: False)...
[2026-02-20 08:11:32] [INFO]  --- END: Windows OS Updates [SUCCESS] ---
```

Cron Job Scheduling Strategy

Cron is used extensively across Linux systems to schedule automated tasks at defined intervals, ensuring that maintenance activities, monitoring checks, and data collection occur reliably without manual intervention. Typical cron jobs include nightly backup verification scripts, hourly certificate expiration checks, daily vulnerability scan initiation, periodic log archival and rotation, and regular system health assessments. Cron jobs are documented with inline comments explaining purpose, frequency, and dependencies, and critical jobs send success/failure notifications to the centralized Discord alerting system to ensure failures are promptly identified and addressed.

Webserver Crontab

Minute	Hour	Day	Month	Weekday	Command	Description	Last Run Timestamp
0	2	*	*	5	/usr/local/bin/upgrade.sh	Weekly system upgrade	Fri, Nov 1, 2025 02:00 AM
30	1	*	*	6	/usr/local/bin/backup_web.sh /var/www/lab /backup	Weekly web backup via rsync	Sat, Nov 2, 2025 01:30 AM

- The first job runs every Friday at 2:00 AM and executes upgrade.sh with no arguments.
- The second job runs every Saturday at 1:30 AM and executes backup_web.sh with two arguments: /var/www/lab(Source directory) and /backup(Target directory).

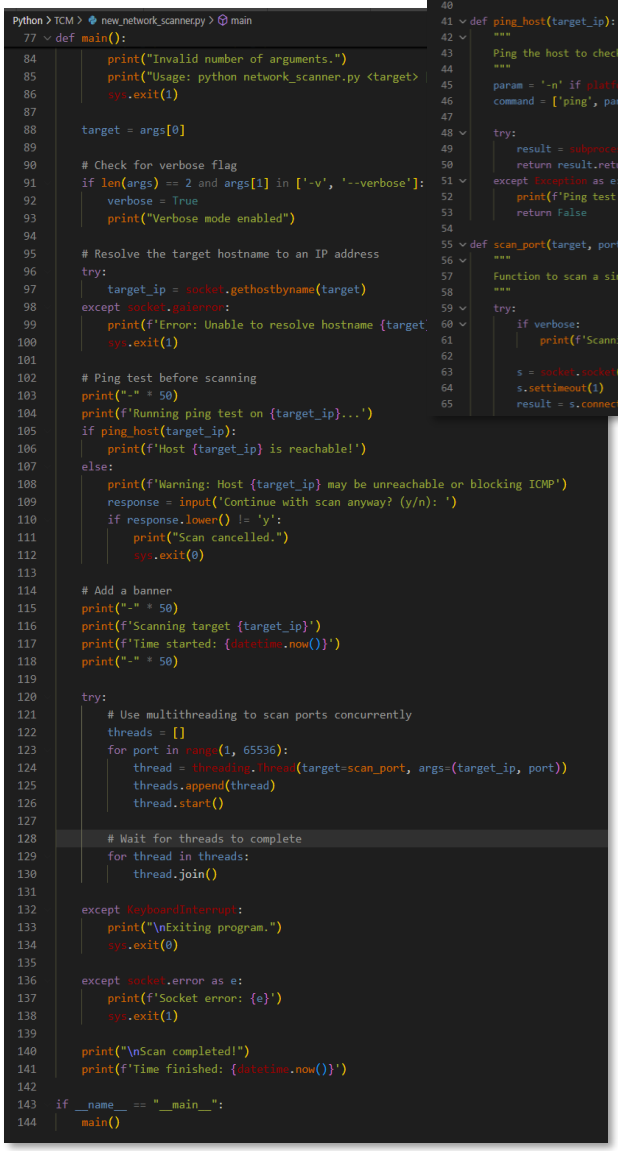
Python Scripting for Advanced Automation

Python scripts are deployed where more complex logic, data processing, or API interaction is required. Python's extensive library ecosystem makes it ideal for tasks like parsing and correlating log data, interacting with REST APIs for service configuration, processing vulnerability scan results, generating custom reports from multiple data sources, and implementing custom security tools. Python's cross-platform nature allows scripts to run consistently across Windows and Linux systems, and virtual environments ensure dependency isolation and reproducibility.

Python Script Example: Network Scanner

A custom Python-based network scanner has been developed to provide tailored reconnaissance capabilities specific to the lab environment. The scanner performs several functions:

- Host Reachability Testing — Pings target hosts before attempting port scans to verify they are online, reducing wasted scan time and providing quick network inventory validation.
- Port Scanning — Uses multi-threaded socket connections to rapidly identify open TCP ports across the full port range (1-65535) or targeted subsets based on scan objectives.
- Service Identification — Labels discovered open ports using a custom service mapping file (common_ports.txt) that includes both standard services (SSH, HTTP, HTTPS) and lab-specific applications (Proxmox, Traefik, Authentik). This makes scan results immediately actionable by identifying what application is likely running on each open port.



Script Integration & Orchestration

Scripts are integrated into broader workflows through several mechanisms. Some scripts are triggered directly by cron jobs for time-based execution. Others are called by monitoring systems in response to alerts or threshold violations (for example, a Grafana alert triggering a remediation script). Ansible playbooks orchestrate multi-step automation by calling shell scripts and Python utilities in sequence, passing parameters and handling error conditions. This layered approach combines the flexibility of custom scripts with the orchestration capabilities of configuration management tools, enabling sophisticated automation scenarios while maintaining maintainability and reusability.

Automation Security Controls

Control Framework:

Control Domain	Implementation	Coverage
Secrets Management	Ansible Vault + Vaultwarden	All credentials encrypted
Access Control	SSH keys only; sudo with NOPASSWD	Ansible service account
Code Integrity	Git version control with commit signing	All IaC assets
Privilege Escalation	Least privilege Proxmox service account	Terraform API access
Input Validation	Regex validation in scripts	Prevents injection attacks
Audit Logging	Syslog + Git commits + n8n execution logs	Full audit trail
Change Management	Git	Controlled deployments
Backup & Recovery	GitHub + NAS + PBS	Multiple restore points

Authentication & Authorization:

Technology	Authentication Method	Authorization Scope
Terraform	Proxmox API token (non-password)	VM/LXC provisioning only
Ansible	SSH key (ed25519; passphrase)	sudo NOPASSWD on managed hosts
n8n	Authentik SSO (OAuth2)	Workflow admin access
GitHub	Personal access token (PAT)	Repository push/pull
Scripts	User context (ansible/root)	File system and service control

Secrets Protection:

Secret Type	Storage Location	Encryption Method
Terraform API tokens	terraform.tfvars (gitignored)	File permissions 0600
Ansible passwords	ansible-vault encrypted files	AES-256 with master key
n8n credentials	Vaultwarden database	Application-level encryption
SSH private keys	~/.ssh/ with 0600 perms	Passphrase-protected
Script API tokens	Environment variables	Not persisted to disk

Audit Trail Components:

Event Type	Logging Mechanism	Retention
Git Commits	"GitHub repository"	Indefinite
Terraform Apply	"Local state file + stdout log"	"90 days"

Ansible Playbook Runs	"Ansible log + syslog"	"90 days"
n8n Workflow Executions	"PostgreSQL + execution history"	"30 days"
Script Executions	"Syslog + individual log files"	"30 days"
Cron Job Runs	"/var/log/cron + job-specific logs"	"30 days"

Operational Resilience & Disaster Recovery

Infrastructure Rebuild Capability:

The entire lab infrastructure can be rebuilt from code repositories in the event of catastrophic failure. Recovery process follows these phases:

Phase 1: Bootstrap Proxmox Host (Manual)

Duration: ~30 minutes

Steps:

1. Install Proxmox VE on bare metal
2. Configure network interfaces and storage
3. Deploy Terraform controller LXC (manual creation)
4. Clone GitHub repositories

Phase 2: Provision Core Infrastructure (Automated)

Duration: ~45 minutes

Steps:

5. Terraform provisions all VMs and LXCs
6. Ansible bootstrap playbook configures SSH access
7. Ansible hardening playbook applies security baseline
8. Ansible service playbooks deploy applications

Phase 3: Restore Data and Configurations (Semi-Automated)

Duration: ~60 minutes

Steps:

9. Restore database backups from PBS/NAS
10. Restore application configs from GitHub
11. Restore secrets from Vaultwarden backup
12. Regenerate certificates from Step-CA

Total Recovery Time: <3 hours (vs. weeks for manual rebuild)

Backup Strategy for Automation Assets:

Asset Type	Backup Method	Frequency	Location
Git Repositories	GitHub remote	On every push	Cloud (GitHub)
Terraform State	rsync to NAS	Daily	NAS + USB
Ansible Inventory	Git versioned	On change	GitHub
n8n Workflows	PostgreSQL dump	Daily	NAS
Custom Scripts	Git versioned	On change	GitHub

Cron Configurations	Ansible playbook managed	On change	GitHub

Monitoring and Alerting:

Metric Monitored	Tool	Alert Threshold	Notification
Terraform apply failures	Exit code	Non-zero exit	Discord
Ansible playbook failures	Return code	Any task failure	Discord
n8n workflow errors	Built-in	Execution failure	Discord
Cron job failures	Syslog	Exit code != 0	Discord
Script execution time	Custom	>2x baseline	Discord

Practical Use Cases and Workflows

Scenario 1: New VM Provisioning

Objective: Deploy new Docker host in <5 minutes

Workflow:

- Developer updates Terraform variables:
  - hostname: docker-vm-03
  - vmid: 203
- Run Terraform:
  - cd terraform/vm
  - terraform apply -var="hostname=docker-vm-03" -var="vmid=203" -auto-approve
- Terraform clones ubuntu-cloud template, provisions VM
- Output displays IP address: 192.168.100.203
- Run Ansible bootstrap:
  - ansible-playbook -i hosts.yml new_install.yaml --limit docker-vm-03.home.com
- Ansible configures SSH, creates ansible user, installs packages
- VM ready for application deployment

Result: Consistent, reproducible VM provisioning with full audit trail

Scenario 2: Weekly Security Update Cycle

Objective: Keep all systems patched without manual intervention

Workflow:

- Saturday 2 AM: n8n workflow triggers
- n8n executes SSH command on Ansible controller
- Ansible playbook runs across all Linux hosts:
- Debian/Ubuntu: apt update && apt upgrade -y
- RHEL/Fedora: dnf update -y
- Playbook output converted to JSON
- JSON uploaded to GitHub (audit trail)
  - Discord notification sent with summary:
  - 23/25 hosts updated successfully
- 2 hosts require reboot
- Admin reviews notification and schedules reboots if needed

Result: Automated patch management with centralized reporting and alerting

Scenario 3: Rapid Disaster Recovery

Objective: Rebuild compromised VM from code

Workflow:

- Incident detected: VM compromised via vulnerability
- Admin destroys compromised VM:
  - terraform destroy -var="vmid=205"
- Review Git history to find last known-good configuration:
  - git log terraform/vm/main.tf
- Checkout previous commit if needed:
  - git checkout abc123def terraform/vm/main.tf
- Rebuild VM with Terraform:
  - terraform apply -var="hostname=web-vm-01" -var="vmid=205"
- Re-run Ansible playbooks to restore configuration:
  - ansible-playbook -i hosts.yml site.yml --limit web-vm-01.home.com
- Restore application data from NAS backup
- VM back online with clean configuration

Result: Complete rebuild in <30 minutes vs. hours of manual work

Scenario 4: Threat Intelligence Distribution

Objective: Daily security awareness for lab operations

Workflow:

- Daily 8 AM: n8n workflow executes
- RSS feeds polled from multiple cybersecurity sources
- Each feed limited to 5 most recent entries
- NIST feed sent to ChatGPT for AI summarization
- Aggregated digest posted to Discord #threat-intel channel

Result: Consolidated threat intelligence reduces information overload

Scenario 5: Configuration Drift Detection

Objective: Ensure compliance with security baseline

Workflow:

- Weekly: n8n triggers Ansible audit playbook
- Playbook gathers configuration from all hosts:
  - SSH config settings
  - Firewall rules
  - Installed packages
  - User accounts
  - DNS configuration
- Output compared against baseline in Git
- Drift detected on 2 hosts (unauthorized package installed)
- Discord alert sent with details

Result: Proactive detection of configuration drift and unauthorized changes